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Soybean Yield Losses Related to Drought Events in Brazil: Spatial–Temporal Trends over Five Decades and Management Strategies

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Abstract: By the end of the decade, the world population is expected to increase by nearly one billion people, posing challenges to meeting global food demand. In this scenario, soybean production is projected to increase by 18% within this decade. Despite being the largest soybean producer, responsible for over 40% of soybeans produced worldwide, drought events often impair Brazilian production. The goals of the present research were to quantify soybean yield losses related to drought in Brazil from 1973 to 2023 at national, state, and municipal levels and to assess the spatial distribution of losses across the production areas. The hypothesis investigated is that year-to-year variations in soybean yield are closely related to water availability, considering that crop management practices are constant from year to year, while increments in soybean yield across time (more than five years) relate tightly to better crop management practices and breeding improvements. Thus, quantifying year-to-year yield losses might demonstrate the effects of water availability on soybean yield. Yield data from the 1976/1977 to 2022/2023 crop seasons from the 26 states and the Federal District came from the National Supply Company, while the Brazilian Institute of Geography and Statistics supplied yield data for the 1973/1974 to 2020/2021 crop seasons from 1998 municipalities with more than 14 crop seasons. Soybean drought yield losses were calculated for each cropping season individually at the municipal, state, and national levels, based on the deviation in the observed yield to the corresponding maximum yield in the five-year window, considering that crop management practices and genetics represent a regular increment in soybean yield, which means that production practices improved over time and deviations from year to year are mainly related to drought occurrence. Annual soybean yield loss (expressed in tons, USD, and percentage), frequency of yield loss, and severity of yield loss were calculated at national, state, and municipal levels for each cropping season. The Standardized Precipitation Index (SPI), acquired from the Brazilian Weather Forecast and Climate Studies Center at the National Space Research Institute, was used as a qualitative indicator to corroborate the assessed soybean yield losses related to drought. The results demonstrate yield losses in more than 50% of crop seasons at the national level, with a similar frequency across the five decades, albeit with lower severities in the last 30 years. The Central–West region was more stable than the South region, with yield losses of up to 74%. In five decades, yield losses related to drought events stand at 11.65%, corresponding to 280 million tons or USD 152 billion (considering the average soybean price in 2022 at the Chicago Board of Trade). At the municipal level, analogous behavior was observed across time and space. The outcomes from the present research might subsidize public and corporative policies related to agricultural zoning, farm loan programs, crop insurance contracts, and food security, contributing to higher agricultural, environmental, economic, and social sustainability.

Keywords: soybean yield and production; growing regions; yield losses; dryness; standardized precipitation index; sustainability

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1. Introduction

By the end of the decade, the world population is expected to increase by nearly one billion people, reaching 8.5 billion in 2030, representing an annual growth rate of 0.9% [1]. This population growth poses challenges to meeting the sustainable development goals of the Food and Agriculture Organization of the United Nations, especially in food security and environmental protection policies. Therefore, sustainable practices in agricultural production play a decisive role in increasing crop yields, meeting global food demands, and, at the same time, protecting the environment and minimizing the pressure on converting non-agricultural lands into croplands [2].

In this scenario, soybean production is projected to increase by 18% within this decade [3], maintaining its participation in food production and serving as one of the main bases of diet for humanity [4]. Brazil is the largest soybean producer, responsible for over 40% (154 million tons) of soybeans produced worldwide (370 million tons) [5,6], and within the next decade, it has been predicted that it will expand its participation in the soybean trade market, becoming responsible for 60% of soybean global exports [3].

Although soybean has been cultivated in Brazil for the last 140 years, Brazilian soybean production only experienced substantial growth in the decade of the 1970s, especially in southern Brazil, reaching 10% of total global production. From the 2000s, with the development of soybean cultivars adapted to tropical conditions, the Central–West region in Brazil became, aside from the southern states, one of the most significant soybean production areas in Brazil. Currently, agricultural areas devoted to soybean production represent about 5% of the Brazilian territory [5]. Through the adoption of soybean production technologies [7], Brazil saw its soybean yield increase from 1500 kg ha^{-1} to 3500 kg ha^{-1} within five decades, which prevented about 60 million hectares from being converted into soybean fields to reach the same production amount [8]. However, despite being a substantial producer, drought periods constantly affect soybean yields in Brazil.

Extreme climate events have a direct impact on social and economic activities, especially global food security. There are indications that drought events in the future will have a higher frequency, longer duration, and greater severity [9–13]. This trend might be especially harmful to Brazil, where 95% of agricultural production depends on rainfall [14].

Drought can be related to lower amounts of precipitation, higher rates of transpiration, lower levels of groundwater, or changes in soil cover, leading to agricultural yield losses, soil degradation, and alterations in ecosystems and water use for human activities [15]. In addition, precipitation distribution through time plays a role in drought characterization, especially for soybean crops, since water deficit during the reproductive stages has a higher impact on yields than water deficit during the vegetative phase [16].

Several authors have characterized drought events as different categories, such as meteorological, climatological, hydrological, agricultural, and socioeconomic [11,17–19]), indicating that insufficient water availability, caused by lower amounts of precipitation and increased water demand, characterizes drought. The interval between a rainfall event and water availability might vary in time and space, making the relationship between water use and its source a highly complex task. According to Cunha et al. [13], the beginning and end of a drought event are difficult to determine, and agriculture is one of the first sectors affected. The agricultural drought definition considers short time intervals related to soil moisture levels insufficient to support crop development. Tucker and Choudhury [20] defined agricultural drought as a time interval with lower plant growing rates caused by lower precipitation volumes regarding the historical average. For Santini et al. [21], drought frequency, duration, severity, and timing are crucial for understanding its impact on agricultural production.

Drought conditions might vary among distinct regions, their climate characteristics, and water use [22], which makes agricultural production a high-risk activity since the entire production chain depends on the climate, one of the most demanding factors to manage and one of the most limiting to maximum yields [23]. In this context, Brazil has a continental territorial extension and a high climatic variability influencing agricultural activities directly [24,25]. According to the Brazilian National Water and Sanitation Agency [26], since 2012, an intense reduction in precipitation indices in different areas of Brazil has been observed.

Understanding the causes of these climatic alterations and precipitation dynamics from year to year is still imprecise due to the short period of observations of such anomalies. Leng and Hall [27] assessed the impacts of drought on crop yield at the global level and emphasized that assessing yield losses exclusively through drought events requires the monitoring of parameters such as water temporal dynamics across key phenological stages, soil water-holding capacity, sowing dates recommended for each region, precipitation, temperature, vapor pressure deficit, solar radiation, and specific crop management practices. Given such limitations, Santini et al. [21] explored yield losses for major crops globally and described that assessing within-country yield loss variability is limited when performing large-scale monitoring. Hence, the impairment of such assessment is due to the variability in edaphoclimatic characteristics of different environments for soybean production in Brazil and the poor distribution of weather stations across the territory, which denote the need for alternative methods for large-scale monitoring of agricultural droughts [13,14]. While Cunha et al. [13] performed a long-term monitoring of extreme drought events in Brazil, demonstrating their differential effects among Brazil's regions under increasing frequency of droughts, Assad et al. [14] developed an integrative review on the resilience of Brazilian agricultural systems due to local climate variability.

Nguyen et al. [28] investigated the corn and soybean yield loss in the US due to drought, demonstrating the importance of climate indices to characterize the water availability to plants. According to the authors, not only the water input (precipitation) should be taken into account but also the water consumption (evapotranspiration). Lu et al. [29] also performed a study on crop yield loss in the US related to drought and used a climate index as a proxy for precipitation monitoring. The authors reinforced the importance of considering the timing, duration, and severity of drought events when investigating crop production. According to Nguyen et al. [28] and Lu et al. [29], when assessing drought effects over crop yield, it is mandatory to detrend the temporal dataset, normalizing the yield improvements obtained through better genetics and crop management practices across time. Based on their results, while improvements in plant breeding and crop management practices are not linear in time and space, year-to-year yield variability is related to drought occurrence.

The goals of the present research were to quantify soybean yield losses related to drought in Brazil from 1973 to 2023 at national, state, and municipal levels and to assess the spatial distribution of losses across the production areas. The hypothesis investigated is that year-to-year variations in soybean yield are closely related to water availability, considering that crop management practices are constant from year to year, while increments in soybean yield across time (more than five years) relate tightly to better crop management practices and breeding improvements. Thus, quantifying year-to-year yield losses might elucidate the effects of water availability on soybean yield.

To the best of our knowledge, this is the first assessment considering both the spatial and temporal aspects of soybean production in the largest global producer. The outcomes of the present research might assist public and corporate policies that drive soybean production in Brazil. These policies have the potential to be adjusted to seek strategies to minimize possible yield losses in future crop seasons, which will result in higher yields and, consequently, higher volumes produced, guaranteeing a food supply based on Brazilian soybeans. Assessing the spatial–temporal trends in yield loss will contribute to enlarging the sustainability of the soybean production chain across agricultural, environmental, economic, and social axes, expressed by lower losses and consequently higher yields; preventing non-agricultural areas from being converted into soybean fields; providing greater profitability and public and corporate policies for the trade market, farm loan programs, and crop insurance contracts; and strengthening food security in Brazil and globally.

2. Materials and Methods

2.1. The Role of Air Temperature, Precipitation, and Solar Radiation in Soybean Yield in Brazil

Precipitation, temperature, and photoperiod are the climatic factors that most influence soybean development and yield [30]. Among these factors, precipitation is the most variable in space and time. Soybean crops adapt better in regions with air temperatures between 20 °C and 30 °C that have adequate solar radiation for each cultivar, and they require from 450 to 800 mm of precipitation, conditions usually found in all soybean production regions in Brazil [24].

The photoperiod in Brazil tends to increase towards the southern region during summer, which coincides with the soybean crop season, suggesting that potentially higher yields are achievable in this region compared to the Central–West region. However, despite having the highest amount of solar radiation availability across crop seasons, the RS is the state with the broadest variability in soybean yield, mainly due to water-deficit periods. In turn, the states from the Central–West region present higher yield stability.

In this context, the total rainfall across crop seasons is relevant but even more important is the distribution of precipitation in the decisive phases of phenological development. According to Monteiro [31], water deficits at sowing and emergence and the flowering and grain-filling stages are critical for soybean production, and drought at the reproductive phases is more harmful to soybean yield than drought periods at the vegetative phases [16,32]).

Figure 1 presents the municipalities in Brazil with soybean crop in the 2020/2021 crop season according to the Brazilian Institute of Geography and Statistics (IBGE) and the climate map of Brazil according to Köppen climate classification.

As observed, the majority of soybean municipalities are located within the Am, Aw, Cfa, and Cfb according to Köppen climate classification. Alvares et al. [24] addressed the climate map of Brazil and found precipitation amounts ranging from 1000 to 2500 mm per year in these climates where soybeans are mostly cultivated. According to the authors, large variability in the total rainfall can be observed within each climate, ranging from approximately 1000 to 3500 mm (Am), 700 to 2500 mm (Aw), 600 to 3500 mm (Cfa), and 700 to 2000 mm (Cfb). Following the Köppen climate classification, the air temperature in these areas during the soybean crop season is suitable for soybean production.

2.2. Soybean Area, Production, and Yield Datasets

Official data on soybean area, production, and yield from all municipalities and states in Brazil (Figure 2) were gathered to assess yield losses. Yield data from the 1976/1977 to 2022/2023 crop seasons from the 26 states and the Federal District were obtained from the National Supply Company (CONAB), a public company under the Ministry of Agriculture, Livestock and Food Supply of Brazil. Yield data from the 1973/1974 to 2020/2021 crop seasons at the municipal level were obtained from the Brazilian Institute of Geography and Statistics (IBGE). The analyzed time span corresponds to the entire soybean production dataset in Brazil and also to the implementation of soybean crops in food production systems in Brazil (from the 1970s).

While the dataset from CONAB (State and National levels) has soybean production data from most of the states across the time span (47 years), at the municipal level, among the 5563 municipalities in Brazil, 2881 had an official record of soybean production in at least one crop season, and 2469 had an official record of soybean production in the 2020/2021 crop season. From those, we analyzed 1998 municipalities with more than 14 crop seasons (30%). Considering the expansion of soybean crop in Brazil, not all the municipalities or states initiated their cultivation in the same year and, therefore, the analyzed time interval might not be the same. Hence, it is impossible to carry out a pair-based comparison of yield losses between municipalities or states for specific years and, if performed, would lead to large uncertainties. Thus, yield loss assessments were performed individually at the national, state, and municipal levels for each cropping season, considering their corresponding datasets. Afterwards, yield losses at each level from each cropping season were grouped into time spans (decades), enabling the comparison of their trends through time. This means that each sample unit (municipality, state, or national) was compared to itself across the different cropping seasons and the trend in yield losses was finally compared among growing regions.

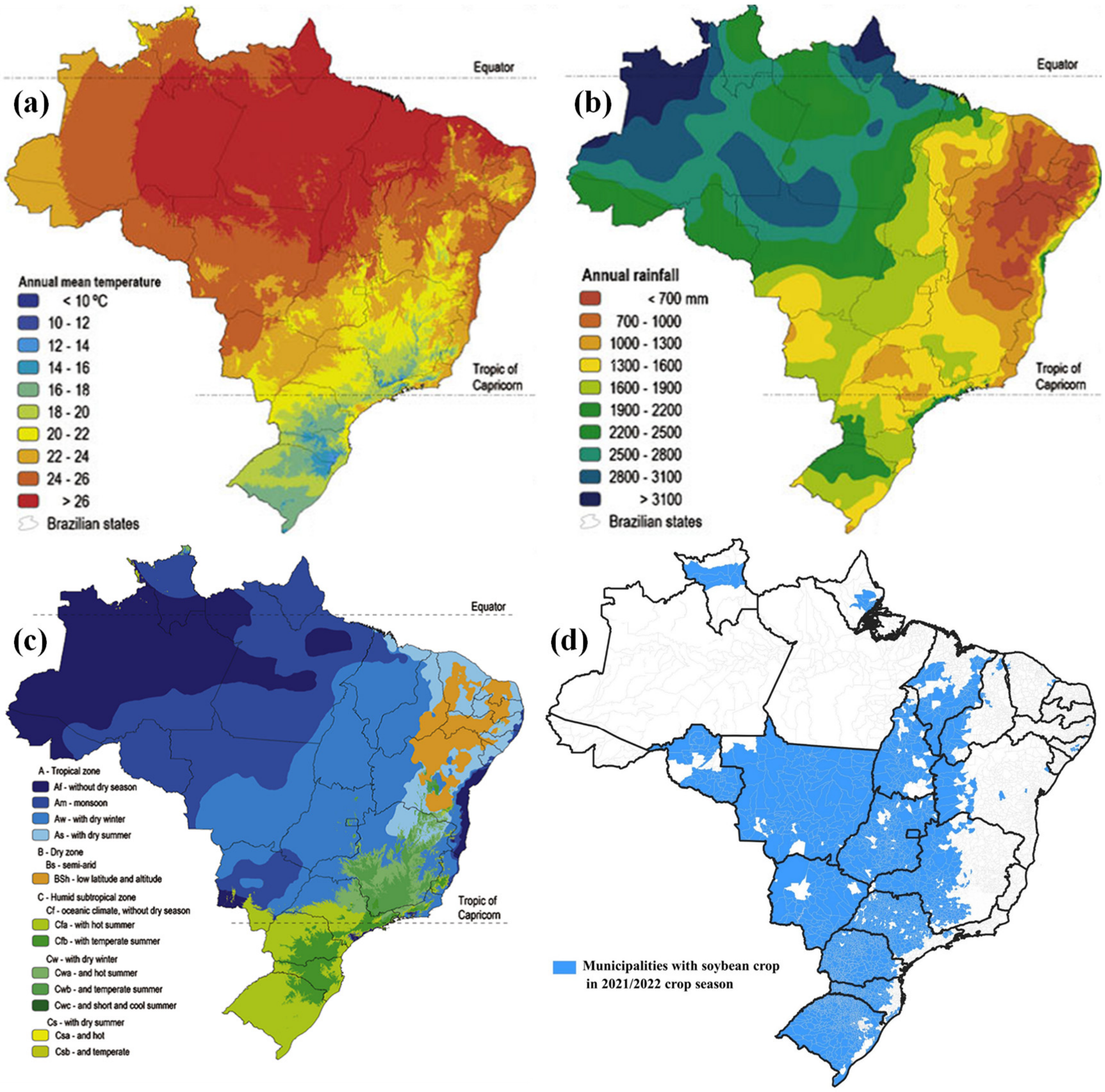

Figure 1. Annual mean temperature (**a**), rainfall (**b**), and climate according to Köppen classification (**c**) for Brazil and municipalities with soybean crops in 2020/2021 crop season (**d**). Subparts a, b, and c were adapted from Alvares et al. [24], and the soybean record from subpart d came from the Brazilian Institute of Geography and Statistics (IBGE).

2.3. Detrending Soybean Yield Time Series and Yield Losses Calculation

Based on the obtained soybean area, production, and yield datasets, soybean yield losses were assessed following the workflow presented in Figure 3.

The following assumptions were the basis for the calculation of soybean yield losses related to drought:

(a) Yield increments through time due to crop management practices and genetic improvements occurred between 1976 and 2023, as demonstrated in Figure 2. Based on historical soybean yield in Brazil, considering crop management practices and genetic improvements regardless of water availability, the national yield increased 45 kg ha^{-1} per year, on average, during the 47 cropping seasons.

(b) Crop management practices and genetics represent a regular increment in soybean yield, which means that production practices improved through time. Such improvements are rarely retrograded. Thus, deviations from year to year are mainly related to drought occurrence. Since yield improved across time, we detrended such improvements by sectioning the yield dataset into five-year moving windows. Hence, we calculated yield loss considering the yield potential within the time a soybean cultivar remains in the market as a highly productive and stable cultivar: 5 years.

(c) Maximum yield in this five-year window was calculated for each of the 1998 municipalities and each state, and also at the national level. This five-year window considered the specific season, two seasons before and two seasons after. The maximum yield in this moving window was assumed as the reference yield. This reference yield represents the maximum yield that could be achieved, individually, for each of the 1998 municipalities and each state, and also at the national level, in a specific year based on available production technologies at the time and the growing region, resulting in year-to-year variability due to water availability.

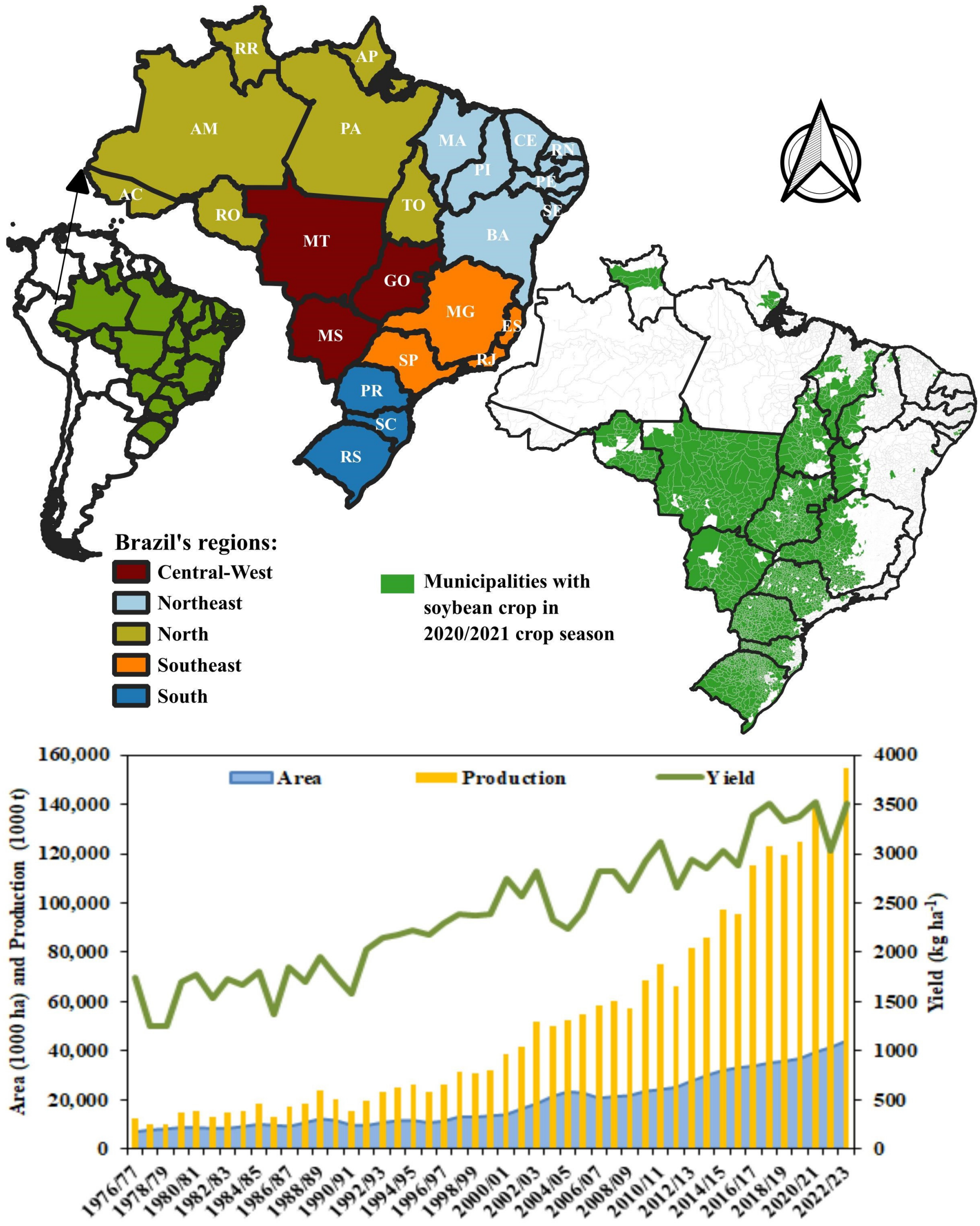

Figure 2. Administrative regions of Brazil, municipalities with soybean crops in 2020/2021 crop seasons, and official data at the national level of soybean area, production, and yield from the 1976/1977 to 2022/2023 crop seasons.

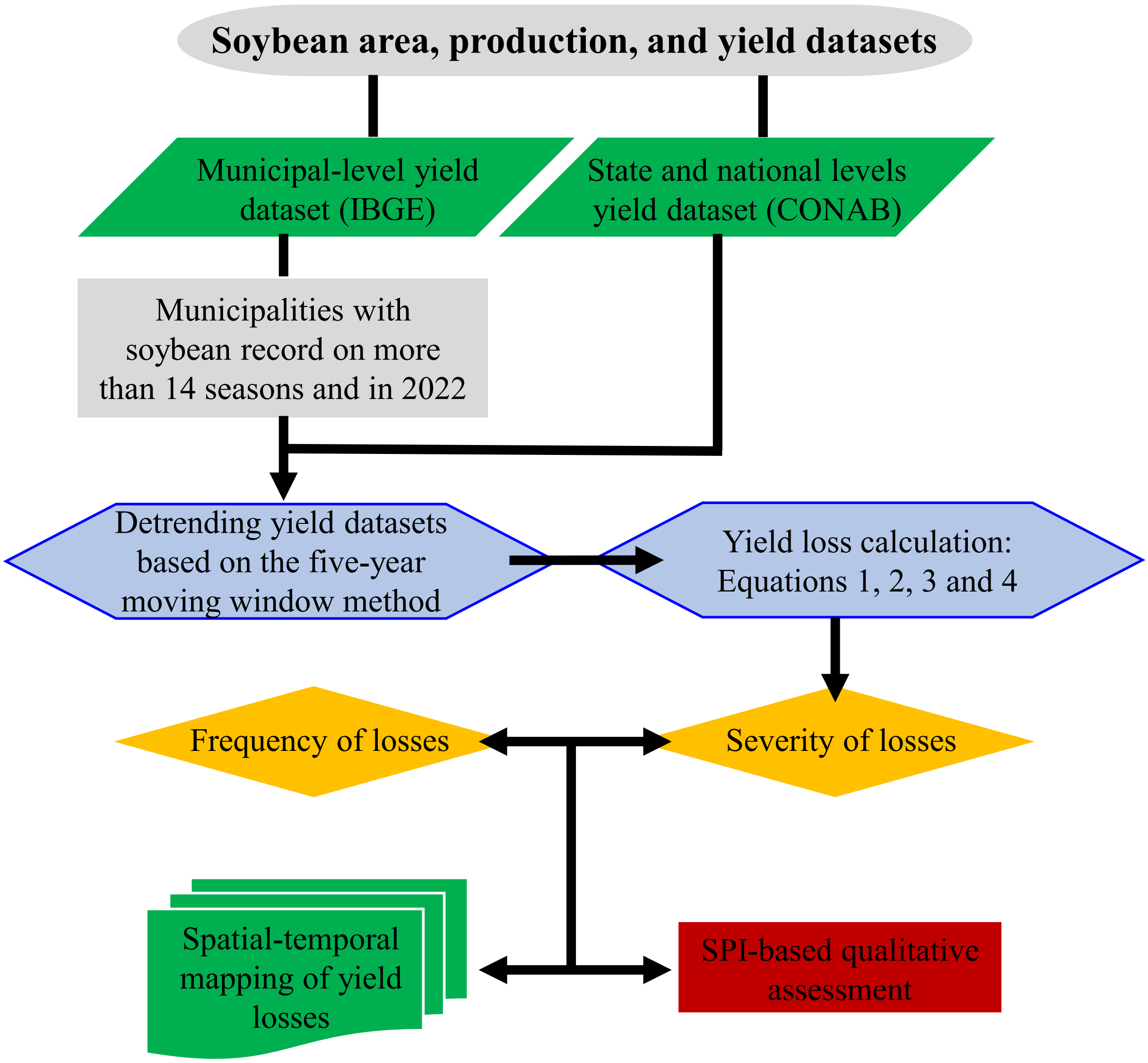

Figure 3. Flowchart of the methodological process.

The year-to-year variability in crop yield has been recognized as likely correlated to drought events by several authors: Hollinger et al. [33], Fernandes and Heinemann [22], and Nguyen et al. [28]. Lu et al. [29] investigated different detrending methods and found the moving window was one of the potential methods for addressing yield losses in the US. In addition, considering the analysis performed individually for each of the 1998 municipalities and each state, and also at the national level, we normalize the variability in crop management practices and production technologies among the different growing regions.

For each crop season, we calculated yield loss individually at the municipal, state, and national levels based on the deviation of the observed yield relative to the corresponding maximum yield in the five-year window, according to Equation (1). This means that for each of the 48 cropping seasons at the municipal level, yield losses were calculated for each of the 1998 municipalities. Similarly, at the state level, yield losses were calculated for each of the 47 cropping seasons for each state. An analogous methodology was applied at the national level.

$$Yield\ Loss = \left(Yield_{Reference} - Yield_{Observed}\right) \tag{1}$$

where *Yield Loss* is the yield loss for a specific year; $Yield_{Reference}$ is the maximum yield in the 5-year window; and $Yield_{Observed}$ is the observed yield for the corresponding year.

Equation (1) can be represented as a percentage of yield loss, according to Equation (2):

$$Yield\ Loss_{Percentage} = \left(1 - \left(\frac{Yield_{Observed}}{Yield_{Reference}}\right)\right) \times 100 \quad (2)$$

To express the yield losses in tons, which means the amount of soybeans that were not harvested due to drought-related events, it is imperative to consider the crop area in each cropping season for each of the 1998 municipalities and each state, and at the national level. Therefore, Equation (1) can be represented in tons according to Equation (3):

$$Yield\ Loss_{tons} = \left(Yield_{\ Reference} - Yield_{Observed}\right) \times \text{crop area} \quad (3)$$

Soybean yield losses from Equation (1) can also be expressed in USD, which means the total income that was not commercialized due to drought-related events, according to Equation (4):

$$Yield\ Loss_{USD} = Yield\ Loss_{tons} \times \text{average soybean price in 2022} \quad (4)$$

The analysis of the dataset of soybean yield losses related to drought events was carried out based on the following steps:

Step (1) Annual soybean yield loss from the 1976/1977 to 2022/2023 cropping seasons at the state and national levels was calculated as the product of yield loss (Equation (1)) and the planted area for the corresponding year and expressed as a percentage (Equation (2)), in tons (Equation (3)), or in USD (Equation (4)), considering the average price for soybeans in 2022 at the Chicago Board of Trade [34].

Step (2) Annual soybean yield loss from the 1973/1974 to 2020/2021 cropping seasons at the municipal level was determined considering only municipalities with at least 14 soybean crop seasons on the period (30% of the 48 crop seasons), which corresponds to 1998 municipalities. Annual soybean yield loss was calculated as the product of yield loss (Equation (1)) and the planted area for the corresponding year and was expressed as a percentage (Equation (2)) or in tons (Equation (3)).

Step (3) The frequency of crop seasons with yield losses at the state and municipal levels was calculated. At the municipal level, the assessment considered only the 1998 municipalities with at least 14 soybean crop seasons during this period (30% of the 48 crop seasons), and the results were expressed as a percentage for the 48 crop seasons (1973/1974 to 2020/2021). At the state level, the assessment comprised the 47 cropping seasons (1976/1977 to 2022/2023).

Step (4) The severity of yield loss at the state and municipal levels was determined. At the municipal level, the assessment considered only the 1998 municipalities with at least 14 soybean crop seasons during this period (30% of the 48 crop seasons). At the state level, the assessment comprised the 47 cropping seasons. The severity of yield loss was grouped as follows: 0–10%, 11–20%, 21–30%, 31–40%, and >40%, and was expressed as a percentage of crop seasons with yield losses.

Spatial datasets of yield losses were generated for the entire territory of Brazil at the state and municipal levels within different time intervals, sectioning by decades and also considering the full-time dataset.

2.4. Standardized Precipitation Index Dataset

We employed the Standardized Precipitation Index (SPI) [17] as an indicator of whether deviations in the Precipitation Index might serve as qualitative indicators of yield loss.

The *SPI* constitutes one of the most-used tools for drought monitoring since adjustment for different time intervals is possible. The *SPI* can also categorize drought in both dry and humid environments and during dry periods or times of excess rainfall, reinforcing its use for agricultural monitoring [35–37]. The *SPI* calculates the deviation of the accumulated precipitation concerning the climatological average for different time scales. Thus, the *SPI* can be summarized as the number of standard deviations of precipitation in relation to historical records (Equation (5)), ideally over a period longer than 30 years. Afterwards, the *SPI* can be calculated for different time spans for cumulated precipitation (e.g., one month, three months, six months), especially for the overall water availability at the regional scale.

$$SPI = \frac{(P - \hat{P})}{\sigma_P} \quad (5)$$

where P is precipitation; $\hat{P}$ is the mean precipitation; and σ_P is the standard deviation of precipitation.

Thus, we acquired monthly *SPI* values from the Brazilian Weather Forecast and Climate Studies Center from the National Institute of Space Research [37] to corroborate the assessed soybean yield loss to drought. The *SPI* calculated from monthly precipitation values had only negative values considered by CPTEC/INPE for drought monitoring. Positive values of *SPI*, for instance, indicate rainfall under normality or above the average. An SPI value equal to zero indicates no abnormal total rainfall for the time span.

Regarding the sowing calendar in most soybean areas in Brazil, we contemplated the accumulated *SPI* only from November, December, and January in the analysis related to the corresponding crop season. The three-month *SPI* was also found to be suitable for a regional assessment of drought by Lu et al. [29] and Nguyen et al. [28]. The *SPI* interpretation followed the methodology for drought monitoring from the National Integrated Drought Information System [38], based on the values determined by the National Climatic Data Center from the National Oceanic and Atmospheric Administration (NCDC/NOAA), as demonstrated in Table 1.

Table 1. Classes of drought intensity based on the SPI [38].

Class	Intensity	Possible Effects	SPI
D0	**Abnormally Dry**	Short-term dryness slowing planting and growth of crops or pastures. Some lingering water deficits. Pastures or crops not fully recovered.	**−0.5 to −0.7**
D1	**Moderate Drought**	Some damage to crops or pastures. Streams, reservoirs, or wells low, with some water shortages developing or imminent. Voluntary water-use restrictions requested.	**−0.8 to −1.2**
D2	**Severe Drought**	Crop or pasture loss likely. Water shortage common. Water restrictions imposed.	**−1.3 to −1.5**
D3	**Extreme Drought**	Major crop/pasture losses. Widespread water shortages or restrictions.	**−1.6 to −1.9**
D4	**Exceptional Drought**	Exceptional and widespread crop/pasture losses. Shortages of water in reservoirs, streams, and wells creating water emergencies.	**−2.0 or less**

The frequencies of drought occurrence from 2003/2004 to 2022/2023 for each Brazilian mesoregion were calculated and expressed as a percentage to meet the official grouping of soybean production areas based on their edaphoclimatic characteristics, which indicates the more suitable cultivars for each mesoregion [39].

Uncertainties on Relying Solely on SPI for Soybean Yield Loss Assessment in Brazil

Droughts are known for inducing crop yield losses and SPI is recognized for matching the rainfall occurrence across key time intervals for crop development, acting, therefore, as a reliable proxy for qualitative assessment of drought occurrence, especially for the overall water availability at the regional scale.

However, some uncertainties arise when addressing the monitoring of soybean yield losses in Brazil, where over 44 million hectares are currently devoted to soybean production. Additionally, the spatial distribution and total area devoted to soybean production significantly increased across the last five decades (which corresponds to the entire soybean production dataset in Brazil and also to the implementation of soybean crop in the food production systems in the country). Such uncertainties can be summarized as follows:

- SPI relies only on rainfall and does not consider, for instance, evapotranspiration, temperature, solar radiation, wind, or the vapor pressure deficit. This means that SPI will represent the amount of precipitation in a specific production area but does not consider the water dynamics in the soil–plant–atmosphere system. Hence, even under lower amounts of precipitation, lower yields may not be fully explained by the SPI if the atmosphere condition has low vapor pressure deficit (highly influenced by temperature). Thus, SPI may provide a regional overview of rainfall but may not represent agricultural drought.
- The time of drought occurrence across the phenological stages plays a crucial role in soybean yield. This means that water deficit induced at the vegetative stages, for instance, may not impair soybean yield, while water deficit at the reproductive stages may cause severe yield impairments. Thus, rainfall distribution may be even more important than the total rainfall. In addition, rainfall in the soybean production areas in Brazil has a large spatial variability within each growing region, which poses challenge to extrapolate the rainfall deviation to all fields within each production area.
- Based on the geographical distribution of soybean crop in Brazil, there are currently up to 2500 cultivars approved by the Ministry of Agriculture, Livestock and Food Supply of Brazil to be implemented in the current cropping season. Most of them are genetically modified organisms (GMOs) and their maturity groups vary from 4 to 9. Among those 2500 cultivars, their cycles may vary, currently, from 100 to 130 days. In addition, the soybean cycle in Brazil has been shortened across time and the growth habit has been adjusted from determinated to indeterminated. This means that across the different cropping seasons in the last five decades, thousands of soybean cultivars with different cycles, different maturity groups, and different adaptation characteristics to the environment have been used.
- Brazil is known for its variety of soil types for food production. Thus, the soybean production fields differ greatly in soil characteristics, mostly those related to the water-holding capacity. This means that two similar growing areas that receive similar amounts of precipitation under similar atmospheric conditions (vapor pressure deficit) may stand up differently to water-deficit periods.
- As a result of the soil type variability in Brazil, soybean producers adopt different levels of management practices. These different practices have been improved across time, but a large variability in agronomic practices and technology for crop production has a significant gap within and among growing regions. This means that soybean fields that adopt higher levels of crop and soil management might present lower yield losses during water-deficit periods.
- On top of the above mentioned uncertainties is the sowing calendar, in Brazil, the Ministry of Agriculture, Livestock and Food Supply has established sowing windows for each growing region in an attempt to achieve higher stability in soybean production. The majority of soybean fields in Brazil are sown from mid-September to mid-December, representing a time span of up to 90 days among the different production areas. In addition, within one growing region, producers may plant their soybean fields on different days, leading to a time span of up to 50 days among them. This means that two similar fields with the same cultivar, same soil type, and same management practices may be sown at different times and, because of this difference, a lower amount of precipitation at a specific time will have different effects on those fields since they will present different phenology.

Based on such uncertainties, a tradeoff arises between the spatial–temporal assessment of drought effects and the in situ crop conditions for yield loss monitoring. Considering that the present research aims to quantify yield losses in Brazil, in which the actual soybean crop area is 44 million hectares, and we aim to perform this assessment across the last five decades, such uncertainties would impair the realistic assessment across time and space. For instance, we need to determine how to empirically validate a method based solely on SPI knowing that the atmospheric conditions may differ significantly in the different growing regions under equal values of SPI and that the soybean fields might have different phenology. If we include the genetic variability and crop and soil management practice differences, we need to identify how to guarantee that all soybean fields from each farm in Brazil within each region will face drought similarly.

To provide an empirical explanation of the limitations of relying solely on SPI for estimating soybean yield loss, we gathered data collected over 11 cropping seasons from field experiments conducted at the experimental farm of the National Soybean Research Center (Embrapa Soja), a branch of the Brazilian Agricultural Research Corporation, located in the municipality of Londrina, Paraná State, Southern Brazil (23°11′03″ and 51°11′37″ S W, 630 m above sea level). Following a split-plot model in a randomized complete block design with four blocks, four water treatments were distributed in the field plots: irrigated (IRR, receiving rainfall and, if necessary, irrigation with a soil water matric potential between 0.03 and 0.05 MPa); non-irrigated (NIRR, receiving only rainfall); water deficit induced in the vegetative stage (WDV); and water deficit induced in the reproductive stage (WDR). Across the ten cropping seasons, 28 genotypes with drought-tolerance genes and different responses to water deficit were evaluated in the subplots. Rainout shelters prevented WDV and WDR plots from receiving rainfall during the vegetative and reproductive phases, contributing to an increase in gravimetric soil moisture variability across treatments. During the period of water-deficit induction in one treatment (WDV or WDR), the other treatment remained watered by rainfall. Shelters automatically covered the plots when the weather station located within the experimental area registered rainfall higher than 0.1 mm, and plots were uncovered when rainfall ceased. Vertical concrete barriers (buried up to a depth of 90 cm) installed around the perimeter of the plots prevented lateral movement of the water from the outside into the plot soil; more details about the field experiment can be found in Crusiol et al. [40]. Weather data (air temperature, relative air humidity, and precipitation) were monitored by the weather station located within the experimental area and SPI was calculated for each experimental treatment from each cropping season based on the precipitation series from Embrapa Soja over 30 years [41]. We adopted the reference yield for each genotype in each cropping season as the yield observed from the IRR treatment (no water restrictions) and calculated yield loss based on Equations (1) and (2).

Figure 4 presents the correlation between SPI and soybean yield in the field experiment located at Embrapa Soja. While in Figure 4a the reference yield was obtained based on the yield under no water restrictions, Figure 4b presents the yield loss considering both the water availability and the variability among soybean genotypes. Considering that all agronomic practices for soybean production are strictly the same among treatments, this limited accuracy in predicting soybean yield loss solely based on SPI is related to precipitation distribution across the cropping seasons (Figure 4a). While plants subjected to water deficit at the vegetative stages of crop development may recover after been re-watered and deliver yields similar to plants under good levels of water availability, water deficit at reproductive stages is known to reduce yield drastically. As can be seen in Figure 4b, in addition to the rainfall distribution, the genetic variability limits the prediction of yield loss based on SPI solely since each genotype has a specific characteristic for drought tolerance. In Figure 4, the negative trend between total rainfall (normalized by the historical record over 30 years) and the soybean yield loss can be observed, demonstrating a general pattern of losses over 30% when SPI values are lower than −1.

Considering the 44 million hectares currently devoted to soybean production in Brazil, solely using SPI for explaining yield loss might be even more challenging considering the complexity from other weather conditions, its spatial variability in small-scale areas (which will certainly drive the water dynamics and affect crop water demand and crop water use), soil types, soil management practices, and, mostly, sowing dates, which are all limitations that still need to be overcome. Therefore, based on the capability of the SPI to express the overall drought occurrence, especially at the regional scale, we adopted an SPI-based analysis for qualitative assessment of water availability. To perform a quantitative analysis of the yield losses spatially and temporally, we used the deviation from the actual yield in a specific crop season (for each municipality and state, and nationally) to the reference yield for this same location within the five-year moving window (detrending the genetic and management improvements across time).

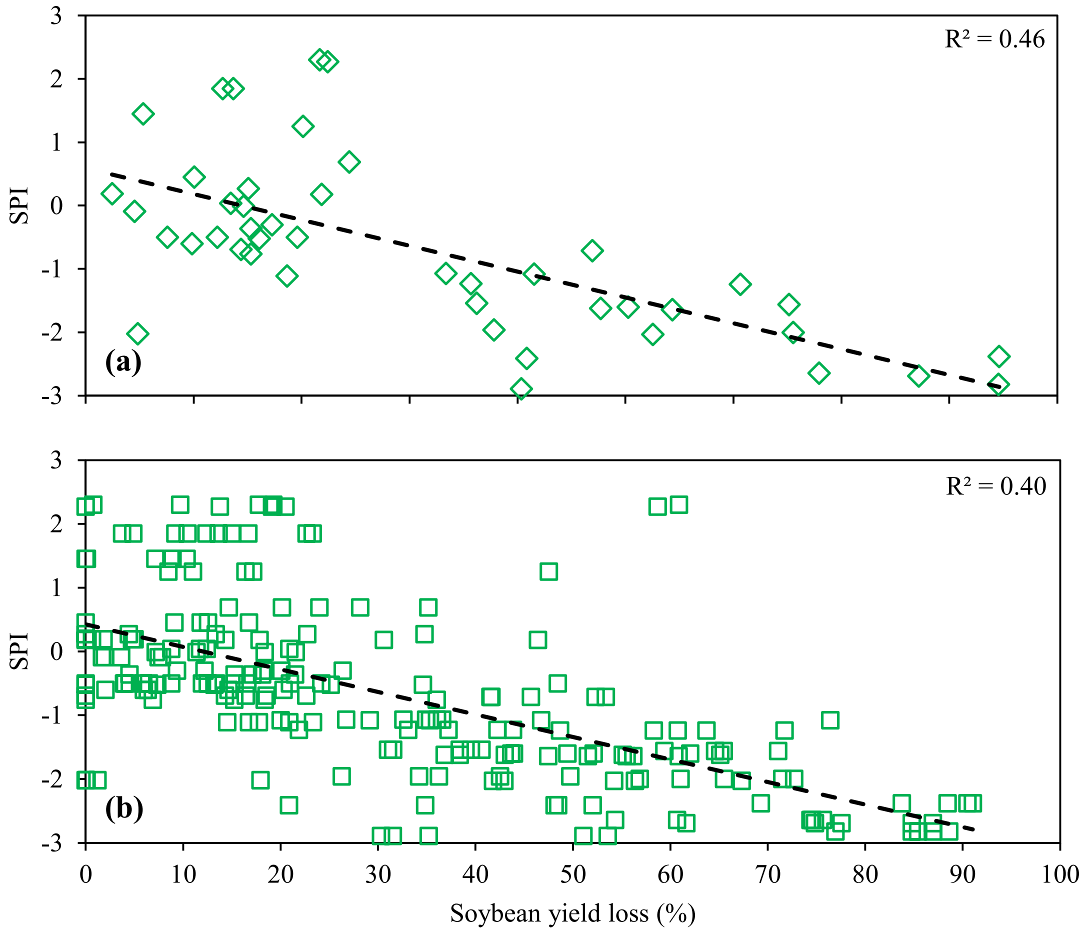

Figure 4. Correlation between SPI and soybean yield in the field experiment located at Embrapa Soja, assuming the reference yield based on the yield under no water restrictions (**a**) and based on genotype performances under no water restrictions (**b**).

3. Results

3.1. Quantification of Soybean Yield Losses at the National and State Level

According to Equation (1), soybean yield losses were initially assessed based on absolute values (tons). However, considering the spatial distribution of soybean production in Brazil, in which each production area has different production amounts, assessing only the absolute values would provide a biased inference. Thus, in addition to assessing soybean yield losses in absolute values (tons) and their equivalence in USD, we also presented the losses in relative values, expressed as a percentage, which enabled comparison among regions over time.

Soybean yield losses, expressed in tons, at the state and national levels from the 1976/1977 to 2022/2023 crop seasons are presented in Table 2, while the corresponding amounts in USD 1000s are in Table 3. The corresponding percentages of yield losses are shown in Table 4.

Table 2. Soybean yield losses, expressed in 1000 tons, at the state and national levels, from the 1976/1977 to 2022/2023 crop seasons.

State and Regions	1976/77 to 1985/86	1986/87 to 1995/96	1996/97 to 2005/06	2006/07 to 2015/16	2016/17 to 2022/23	47 Crop Seasons
North	0.23	44.36	361.19	2299.00	2297.87	5002.65
RR	-	0.14	10.50	29.46	41.49	81.59
RO	0.23	7.79	61.41	116.05	340.05	525.53
AC	-	-	-	-	11.24	11.24
AM	-	-	1.83	-	3.93	5.75
AP	-	-	-	-	7.31	7.31
PA	-	-	16.36	374.05	378.23	768.64
TO	-	36.43	271.09	1779.43	1515.63	3602.58
Northeast	42.38	1732.74	3756.36	12,751.48	4833.88	23,116.83
MA	8.71	94.58	490.42	2211.02	982.51	3787.25
PI	-	8.45	461.84	3174.56	1045.16	4690.01
CE	-	-	-	-	-	-
RN	-	-	-	-	-	-
PB	-	-	-	-	-	-
PE	-	-	-	-	-	-
AL	0.24	0.33	-	-	5.47	6.04
SE	-	-	-	-	-	-
BA	33.43	1629.38	2804.10	7365.90	2800.74	14,633.53
Central–West	2608.29	7718.00	18,488.21	25,483.53	20,701.93	74,999.96
MT	301.99	2965.01	6765.70	11,894.93	9505.04	31,432.67
MS	1737.63	2450.42	6970.42	6411.16	6853.28	24,422.91
GO	531.79	2176.44	4626.10	7026.21	4241.92	18,602.46
DF	36.87	126.14	126.00	151.23	101.68	541.92
Southeast	1311.38	2148.85	3521.94	5324.54	2783.17	15,089.88
MG	420.98	1035.53	1484.43	3164.02	1539.10	7644.06
ES	-	-	-	-	-	-
RJ	-	-	-	-	-	-
SP	890.40	1113.32	2037.51	2160.52	1244.07	7445.82
South	17,175.87	15,559.12	38,215.38	38,900.18	51,768.59	161,619.15
PR	7222.71	4086.34	11,009.52	18,300.33	18,018.21	58,637.10
SC	734.01	813.58	1208.44	1565.66	1548.98	5870.67
RS	9219.16	10,659.21	25,997.41	19,034.19	32,201.40	97,111.37
Brazil	21,138.16	27,203.07	64,343.08	84,758.73	82,385.43	279,828.46

AC: Acre; AL: Alagoas; AP: Amapá; AM: Amazonas; BA: Bahia; CE: Ceará; DF: Distrito Federal; ES: Espírito Santo; GO: Goiás; MA: Maranhão; MT: Mato Grosso; MS: Mato Grosso do Sul; MG: Minas Gerais; PA: Pará; PB: Paraíba; PR: Paraná; PE: Pernambuco; PI: Piauí; RJ: Rio de Janeiro; RN: Rio Grande do Norte; RS: Rio Grande do Sul; RO: Rondônia; RR: Roraima; SC: Santa Catarina; SP: São Paulo; SE: Sergipe; TO: Tocantins.

From 1976/1977 to 2022/2023 (47 crop seasons), soybean yield losses in Brazil reached nearly 280 million tons (Table 2), which corresponds to USD 152 billion (Table 3) and 11.65% during this interval (Table 4). Soybean yield losses demonstrated an increment in tons and USD across time, related to the increment in soybean area (Figure 2). When assessing losses in yield concerning the production percentage, the highest yield losses were observed from 1976 to 1985 and 1996 to 2005 (Table 4).

Specifically, from 1976/1977 to 1985/1986, RO State demonstrated yield losses of 46% (Table 4), which also might be related to the lack of specific crop management practices since, at the time, RO State was a new agricultural frontier. However, the yield loss in RO State from 1976/1977 to 1985/1986 (Table 2) accounted for just 230 tons against 21 million tons accounted from the entire country, which represents only about 0.001% of yield loss at the national level. In addition, during the same time interval, MA State presented an analogous behavior.

It is relevant to highlight that the distribution of soybean yield losses is uneven across Brazil. In the 47 crop seasons analyzed, the North, Central–West, and Southeast regions demonstrated the lowest losses (below 8%), while the Northeast region presented yield losses of around 13%, and the South region exhibited the highest percentage of soybean yield loss: 18%. Although PR and SC States showed yield losses of 13 and 12%, respectively, RS State proved to be more affected by drought periods, with losses of 25% (Table 4).

Table 3. Soybean yield losses, expressed in USD 1000s, at the state and national levels, from 1976/1977 to 2022/2023 crop season.

State and Regions	1976/77 to 1985/86	1986/87 to 1995/96	1996/97 to 2005/06	2006/07 to 2015/16	2016/17 to 2022/23	47 Crop Seasons
North	127	24,216	196,969	1,253,648	1,253,048	2,728,009
RR	-	76	5725	16,114	22,625	44,541
RO	127	4270	33,484	63,282	185,429	286,592
AC	-	-	-	-	6131	6131
AM	-	-	997	-	2140	3137
AP	-	-	-	-	4003	4003
PA	-	-	8921	203,962	206,272	419,155
TO	-	19,870	147,841	970,290	826,447	1,964,449
Northeast	23,110	944,879	2,048,312	6,953,156	2,635,872	12,605,330
MA	4751	51,581	267,463	1,205,637	535,744	2,065,176
PI	-	4635	251,831	1,731,044	569,925	2,557,435
CE	-	-	-	-	1	1
RN	-	-	-	-	-	-
PB	-	-	-	-	-	-
PE	-	-	-	-	-	-
AL	133	178	-	-	3014	3324
SE	-	-	-	-	-	-
BA	18,226	888,486	1,529,018	4,016,476	1,527,188	7,979,394
Central–West	1,422,247	4,208,481	10,081,282	13,895,757	11,288,414	40,896,181
MT	164,668	1,616,762	3,689,200	6,486,097	5,182,921	17,139,647
MS	947,496	1,336,163	3,800,830	3,495,905	3,736,971	13,317,364
GO	289,977	1,186,768	2,522,537	3,831,280	2,313,053	10,143,615
DF	20,107	68,788	68,715	82,475	55,470	295,555
Southeast	715,068	1,171,725	1,920,467	2,903,384	1,517,642	8,228,286
MG	229,551	564,652	809,435	1,725,297	839,253	4,168,188
ES	-	-	-	-	-	-
RJ	-	-	-	-	-	-
SP	485,518	607,073	1,111,032	1,178,087	678,389	4,060,098
South	9,365,661	8,484,106	20,838,097	21,211,552	28,228,413	88,127,828
PR	3,938,397	2,228,224	6,003,283	9,978,817	9,824,969	31,973,691
SC	400,241	443,626	658,944	853,750	844,664	3,201,225
RS	5,027,022	5,812,255	14,175,870	10,378,985	17,558,781	52,952,913
Brazil	11,526,214	14,833,408	35,085,126	46,217,498	44,923,389	152,585,634

AC: Acre; AL: Alagoas; AP: Amapá; AM: Amazonas; BA: Bahia; CE: Ceará; DF: Distrito Federal; ES: Espírito Santo; GO: Goiás; MA: Maranhão; MT: Mato Grosso; MS: Mato Grosso do Sul; MG: Minas Gerais; PA: Pará; PB: Paraíba; PR: Paraná; PE: Pernambuco; PI: Piauí; RJ: Rio de Janeiro; RN: Rio Grande do Norte; RS: Rio Grande do Sul; RO: Rondônia; RR: Roraima; SC: Santa Catarina; SP: São Paulo; SE: Sergipe; TO: Tocantins.

The obtained results evidence the importance of agricultural policies to manage possible losses in the future, considering that PR and RS States, both in southern Brazil, were responsible (in the 2022/2023 crop season) for 20% of Brazilian soybean production and nearly 10% of the soybean produced worldwide [5,6]). In addition, yield losses in the Northeast region are related mainly to the soybean area known as MATOPIBA (initials from MA, TO, PI, and BA States).

3.2. Spatial Distribution of Frequency and Severity of Soybean Yield Losses at the National Level

Despite analyzing soybean yield losses in absolute and relative values (Tables 2–4), we assessed, at the state level, the frequency of crop seasons with yield losses and their severity (Figure 5).

Table 4. Soybean yield losses, expressed as a percentage, at the state and national levels, from the 1976/1977 to 2022/2023 crop seasons.

State and Regions	1976/77 to 1985/86	1986/87 to 1995/96	1996/97 to 2005/06	2006/07 to 2015/16	2016/17 to 2022/23	47 Crop Seasons
North	46.67	10.30	7.00	9.59	4.55	6.25
RR	-	0.65	8.09	8.55	3.14	4.49
RO	46.67	9.21	5.81	2.40	3.60	3.41
AC	-	-	-	-	11.22	11.22
AM	-	-	6.60	-	6.51	6.54
AP	-	-	-	-	2.62	2.62
PA	-	-	2.70	7.57	2.63	3.85
TO	-	11.23	8.12	12.85	6.08	8.49
Northeast	14.75	27.02	15.45	22.90	5.60	13.36
MA	38.56	13.24	8.12	14.94	4.39	8.62
PI	-	13.81	20.09	31.01	5.48	14.81
CE	-	-	-	-	0.01	0.01
RN	-	-	-	-	-	-
PB	-	-	-	-	-	-
PE	-	-	-	-	-	-
AL	8.70	2.10	-	-	12.29	9.61
SE	-	-	-	-	-	-
BA	12.76	28.98	17.56	24.03	6.25	15.03
Central–West	10.66	9.79	9.50	7.22	4.79	6.93
MT	6.51	7.99	6.00	5.50	3.71	5.01
MS	13.25	10.64	21.90	11.63	9.19	12.36
GO	9.10	12.25	9.45	8.82	4.26	7.39
DF	10.63	13.62	11.59	8.63	5.18	8.92
Southeast	10.67	10.04	10.31	10.71	3.84	7.96
MG	10.84	9.91	7.72	9.98	3.40	6.91
ES	-	-	-	-	-	-
RJ	-	-	-	-	-	-
SP	10.60	10.17	13.66	11.98	4.60	9.39
South	18.04	14.93	25.02	14.73	19.86	18.43
PR	17.90	8.67	12.64	13.21	13.55	13.16
SC	13.69	16.19	20.26	11.01	9.33	12.45
RS	18.63	20.48	43.56	17.10	28.98	25.32
Brazil	15.91	12.88	15.66	11.36	9.13	11.65

AC: Acre; AL: Alagoas; AP: Amapá; AM: Amazonas; BA: Bahia; CE: Ceará; DF: Distrito Federal; ES: Espírito Santo; GO: Goiás; MA: Maranhão; MT: Mato Grosso; MS: Mato Grosso do Sul; MG: Minas Gerais; PA: Pará; PB: Paraíba; PR: Paraná; PE: Pernambuco; PI: Piauí; RJ: Rio de Janeiro; RN: Rio Grande do Norte; RS: Rio Grande do Sul; RO: Rondônia; RR: Roraima; SC: Santa Catarina; SP: São Paulo; SE: Sergipe; TO: Tocantins.

In Figure 5a, one can observe that only AP, RR, and PI States registered yield loss occurrences in less than 60% of their soybean historical dataset. In turn, the five largest soybean producers in Brazil (GO, MS, MT, PR, and RS) demonstrated a frequency of yield losses of more than 70% in their crop seasons, with the highest frequencies for the states located in the Central–West region (GO, MS, and MT).

However, most crop seasons with yield losses in the Central–West region had less than 10% of their production impaired (Figure 5b). Yield losses from 10 to 20% showed an even frequency distribution across Brazil, except for RR State (Figure 5c). For the class of 20 to 30% losses (Figure 5d), the highest frequencies were observed in the states of PR, SC, and RS (located in the South region) and BA and PI (located in the Northeast region). We call attention to RS and BA, the states with the highest frequencies of yield losses between 30 and 40%: 10.53% and 11.76%, respectively (Figure 5e). One should note that RS has a production area of 6.5 million hectares against 1.9 million hectares devoted to soybean in BA in the 2022/2023 crop season [5], which makes yield losses in RS State more impactful to the economy and food production than BA.

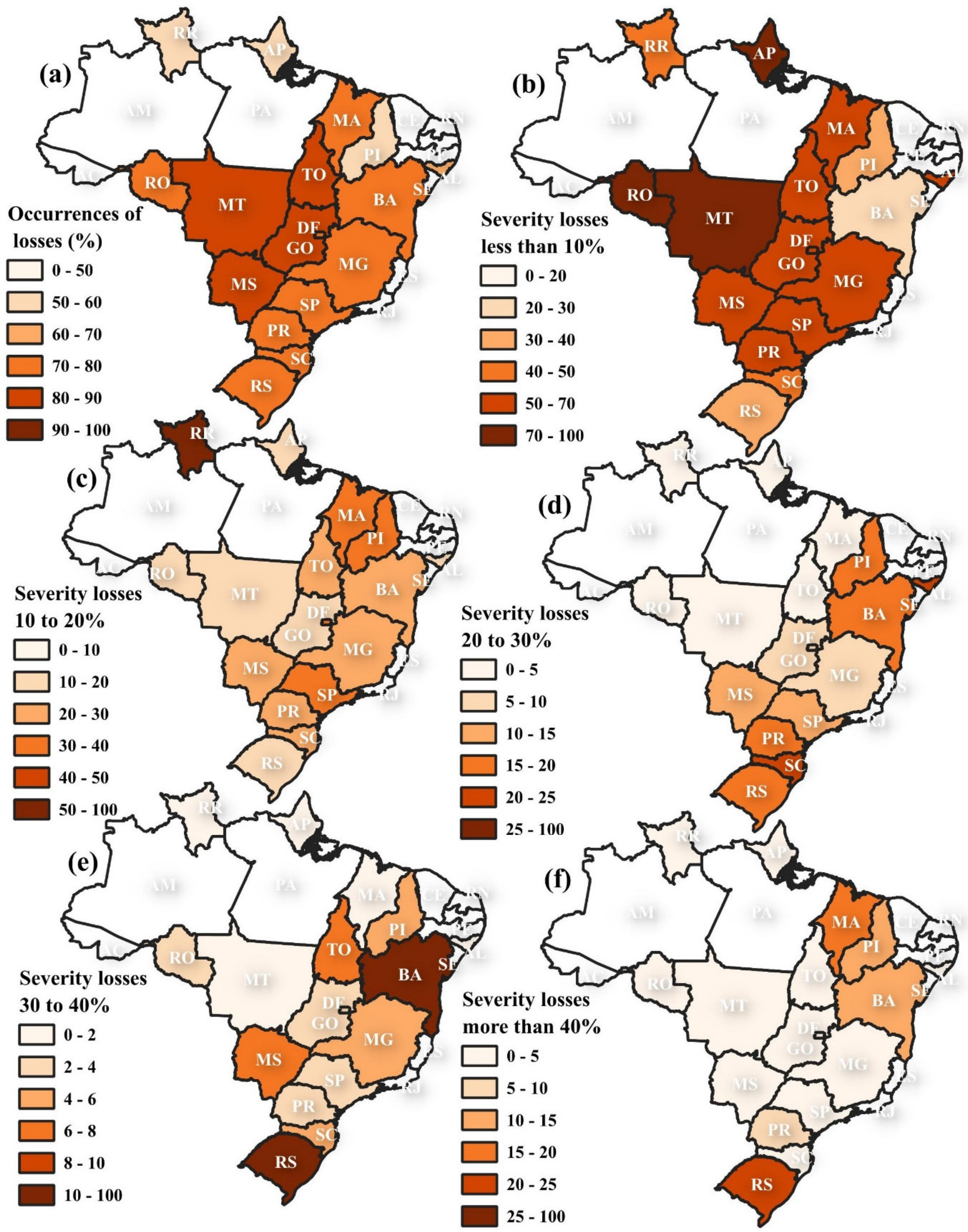

Figure 5. Frequency of soybean yield losses at the state level (**a**) and severity of losses: 0–10% (**b**), 10–20% (**c**), 20–30% (**d**), 30–40% (**e**), and more than 40% (**f**).

Soybean yield losses of over 40% had the highest frequencies of occurrence in the States of MA, PI, and BA (which represent the MATOPIBA soybean expansion area), and PR and RS, representing two of the largest producers in Brazil. Specifically, RS State had over 23% of crop seasons with yield losses over 40%.

As observed in Figure 5, soybean yield losses are distributed unevenly in Brazil. Based on this scenario, national production might be compromised when drought periods affect a significant production region with high severity, especially the South region and specifically the RS State. Thus, considering the uneven distribution of yield losses in frequency and severity, Figure 6 presents the soybean production and yield losses (in both production and percentage) at the national level across the 47 crop seasons.

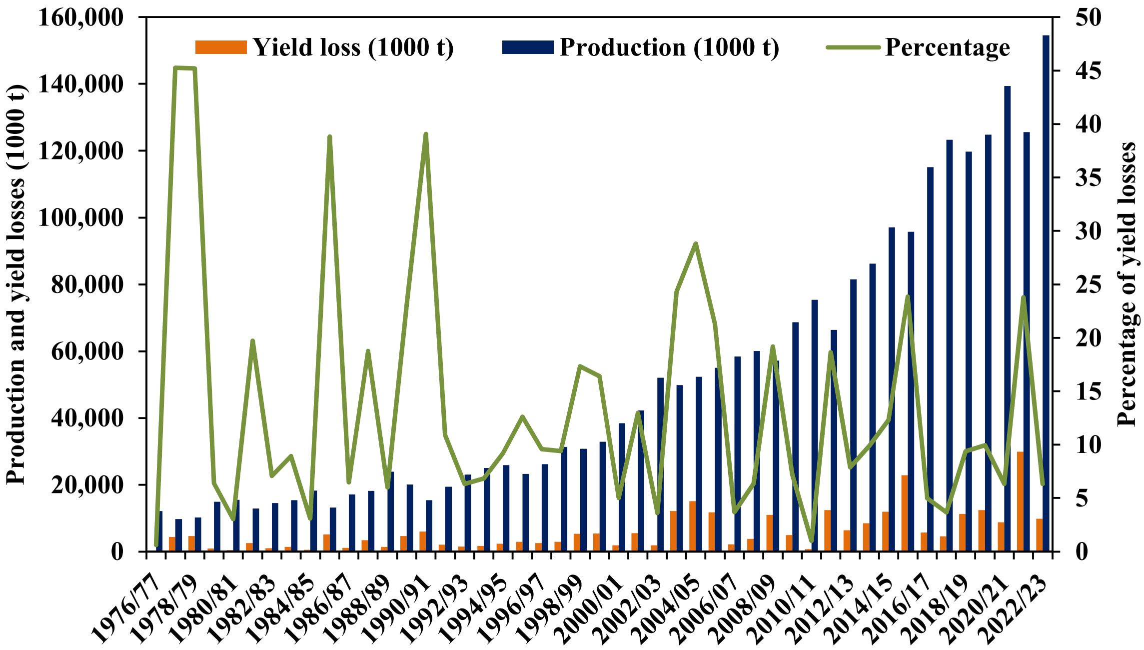

Figure 6. Soybean production and yield losses in Brazil from the 1976/1977 to 2022/2023 crop seasons and the percentage of yield losses in relation to the $\text{Yield}_{\text{Reference}}$ (Equation (1)) for the corresponding crop season.

Figure 6 shows the increment in soybean production and yield losses across the 47 crop seasons. It is worth noting that the frequency of yield losses at the national level presents a similar pattern from 1976/1977 to 2022/2023, but the yield loss severity proved lower from 1990. While yield losses of around 40% occurred in the 1977/1978, 1978/1979, 1985/1986, and 1990/1991 seasons, yield losses of about 20% appeared in the past 15 years: 2008/2009, 2011/2012, 2015/2016 and 2021/2022. This shift is mainly due to better adjustments in crop practices, especially those related to agricultural water management, and also the implementation of public policies based on the agricultural zoning of climatic risk. It is valuable to highlight the three consecutive crop seasons, 2003/2004, 2004/2005, and 2005/2006, when yield losses from 21.3 to 28.8% seriously compromised the soybean production chain in Brazil.

However, as evidenced in Tables 2–4, yield losses are distributed unequally across the soybean areas in Brazil; the same is true of their severity (Figure 5). This trend in frequency, severity, and amount of lost soybean, mainly due to droughts, represents a pressure in regions broadly devoted to soybean production, like the state of RS (Figure 7), demonstrating that severe regional yield losses have a highly negative impact on national soybean production.

An example of such pressure on the state of RS is the 2022/2023 crop season: in that crop season, Brazil had the highest soybean production in history (154 million tons) and a yield loss of about 6.3%. This 6.3% is concentrated mainly in RS State, with yield losses of 9.4 million tons, representing 42.1% of the state yield loss and 97% of the national yield loss in that crop season. In addition, RS, GO, and DF were the only states with yield losses in the 2022/2023 crop season, even though the observed yield loss percentages were minute (1.5 and 0.6, respectively). Such a result is of extreme importance, considering that the biggest soybean crop season in Brazil (2022/2023) could be even larger, achieving over 164 million tons, which would represent an additional input of USD 5 billion in the National economy, with a positive contribution to the global food supply.

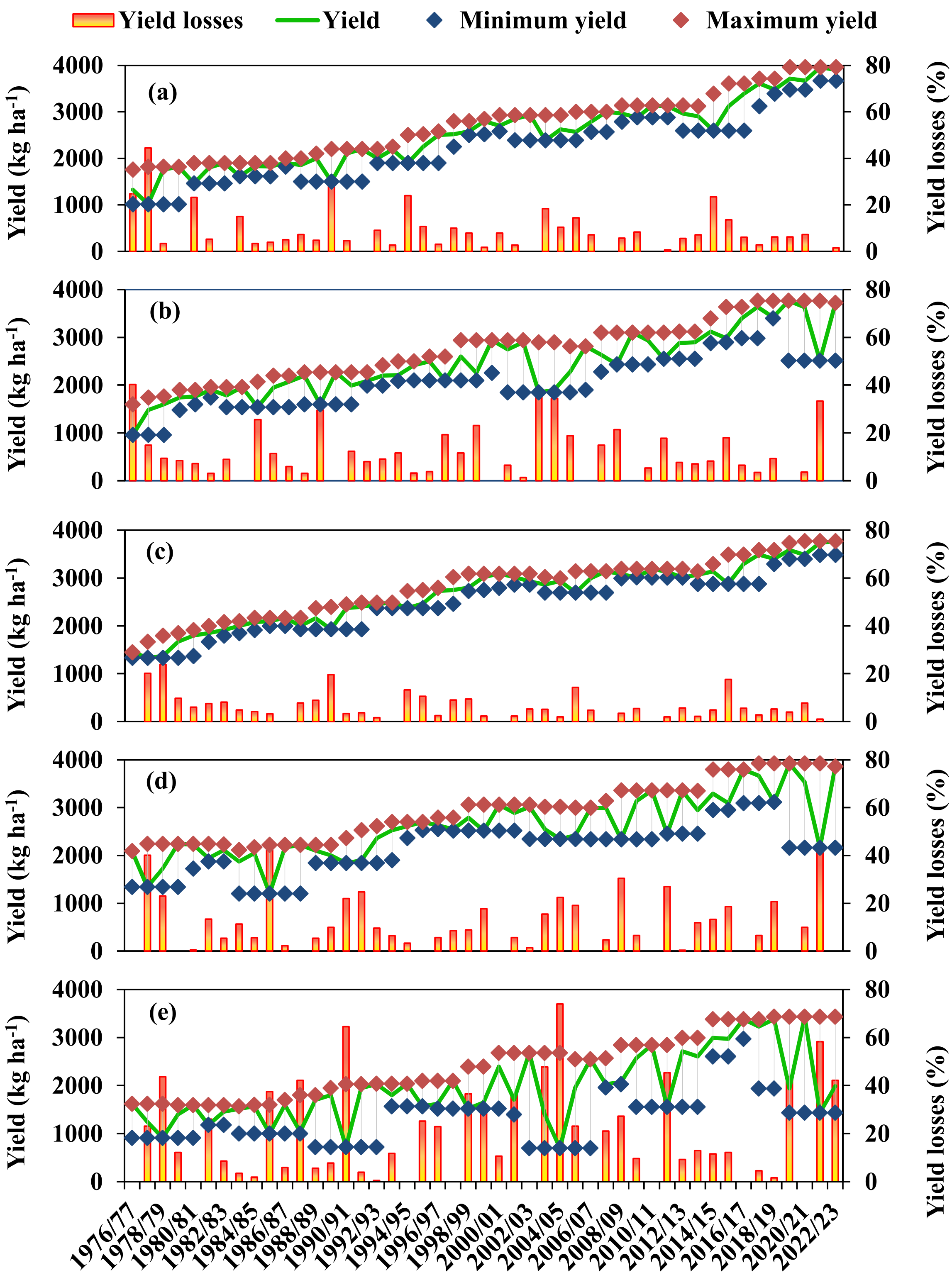

Figure 7. Observed minimum and maximum soybean yields from 1976/1977 to 2022/2023 in five-year windows in the states of GO (**a**), MS (**b**), MT (**c**), PR (**d**), and RS (**e**).

Figure 7 evidences the differential impact of yield losses in five of the largest soybean producers in Brazil: GO, MS, MT, PR, and RS. In all states, soybean yield increments occurred through the 47 crop seasons. These overall increments in yield are related to crop and soil management improvements and better genetics for soybean cultivars.

The larger amplitude between minimum and maximum yield in the five-year window occurred in the southern Brazilian states, PR and RS (Figure 7d,e). The amplitudes observed for the five largest soybean producers in Brazil denote that the states of GO, MS, and MT (Figure 7a–c) are more stable concerning drought occurrence than the states located in southern Brazil.

The state of MT is the largest soybean producer in Brazil. In the 2022/2023 crop season, the state had 12 million ha devoted to soybean crops, resulting in the production of over 45 million tons [5], corresponding to 12% of the soybean produced worldwide [6]. From Figure 7c, it is possible to observe the yield stability in MT, with losses below 20%. Similarly, GO had lower yield loss rates than MS, PR, and RS, albeit with higher rates than MT (Figure 7a). In turn, PR (Figure 7d) presented yield amplitude in the five-year window, leading to yield losses of up to 40%, while RS (Figure 7e) had the largest yield amplitude in the five-year window, with losses of up to 74% (2004/2005 crop season).

In five crop seasons, RS State experienced a sequence of events: 36.4% yield loss (2001/2002); breaking the record for yield and production amount (2002/2003); 47.8% yield loss (2003/2004); 74% yield loss (2004/2005); 23.1% yield loss (2005/2006); followed by another record-breaking year in terms of yield and production (2006/2007). This sequence evidences the yield instability in RS State due to climate conditions.

MS State presented relative yield stability across the 47 crop seasons (Figure 7b). However, some specific years showed yield losses of up to 34% (2004/2005 crop seasons). This trend is related to the municipality assessment of yield losses, which indicates the southeastern regions of the State are most prone to drought occurrence.

3.3. The Standardized Precipitation Index (SPI) and Its Relation with Spatial Distribution of Soybean Yield Losses

Considering that the South region of Brazil is more susceptible to yield losses related to drought, especially the RS State, we selected four cropping seasons with the largest yield losses to perform a qualitative comparison with the climate conditions through those cropping seasons. To corroborate the yield loss assessment related to drought events, Figure 8 presents the SPI from November to January in the 1990/1991, 2003/2004, 2004/2005, and 2021/2022 crop seasons. As observed in Figure 7, these four cropping seasons had yield losses of 64.5%, 47.8%, 74%, and 58.3%, respectively. In the 1990/1991 crop season, a loss of 39.1% was observed at the national level, corresponding to six million tons or USD 3.27 billion. Among the six million tons of soybean production impaired by drought, 93% occurred in the South region (71% only in the RS State, with yield loss of up to 64.5%). Therefore, Figure 8 evidences that the D4 (Exceptional Drought) conditions in RS State in January 1991 seriously compromised the RS State soybean production. Although D1 (Moderate Drought) conditions were also observed in the Central–West region, this area was responsible for only 38% of the soybean crop area at that time. In addition, considering the sowing calendar, the moderate drought observed in December did not match the reproductive phase of soybean development when grains were forming. Grain filling, a soybean development phase critically affected by drought, coincided with the exceptional drought conditions in RS State in January 1991.

Similarly, in the 2003/2004 crop season, SPI values ranging from D2 (Severe Drought) to D4 (Exceptional Drought) were observed in MS, PR, SC, and RS States in January 2004. In this crop season, soybean yield loss was 24.3%, equivalent to 12.1 million tons, or USD 6.6 billion. Specifically, the yield losses at the state level were 36.2% (MS), 15.5% (PR), 25.9% (SC), and 47.8% (RS). In the following crop season, 2004/2005, soybean yield loss in Brazil was 28.8%, representing 15 million tons or USD 8.2 billion. The states where soybean production was particularly impaired were MS (34.4%), PR (22.4%), SC (37.2%), and RS (74%). However, differently from the 1990/1991 and 2003/2004 seasons, the drought, evidenced by the SPI, occurred in December instead of January, corresponding to the beginning of the reproductive phase, which is also critical for soybean production since it corresponds to the flowering stages of crop development.

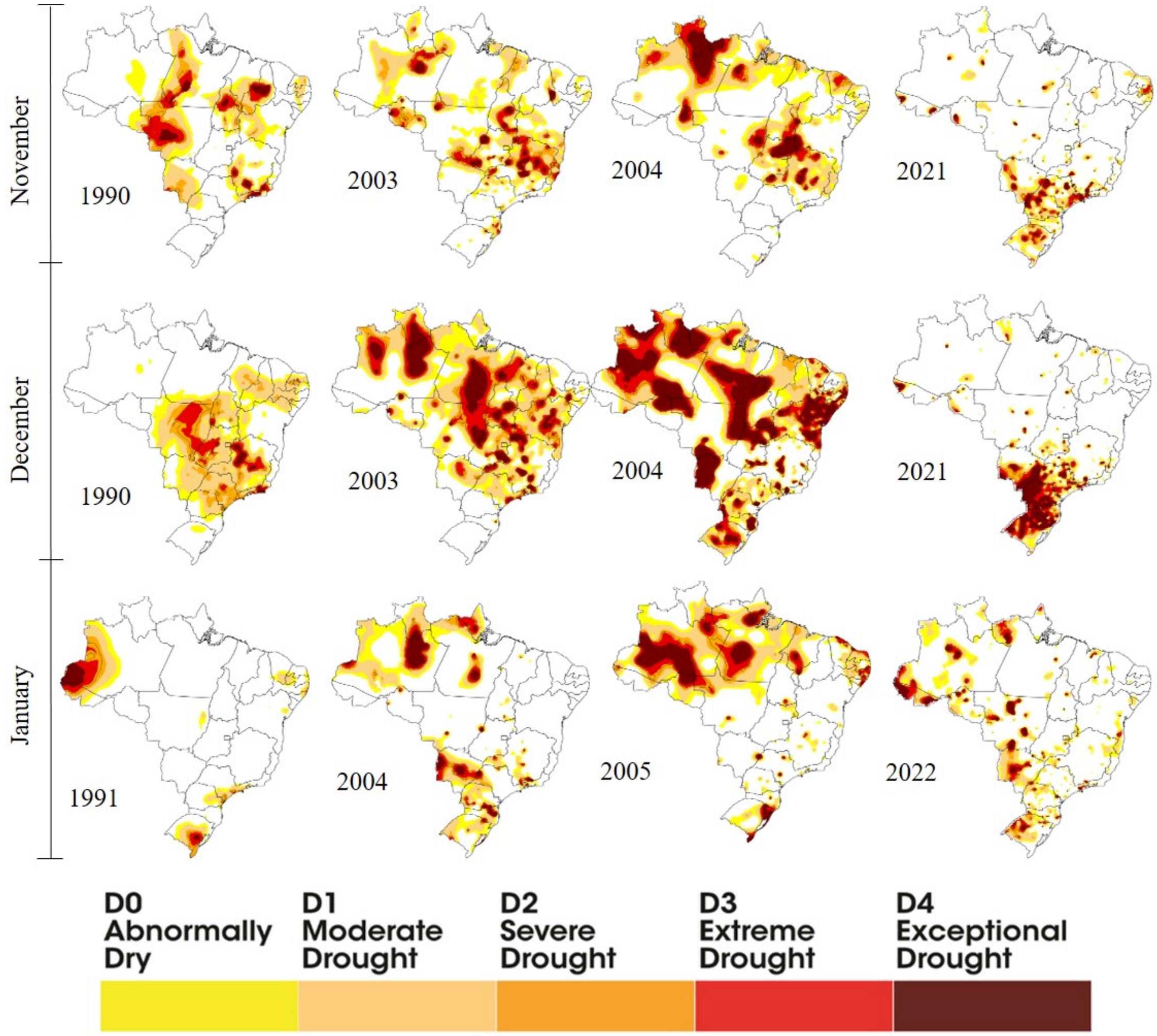

Figure 8. Monthly SPI maps from November, December, and January in the 1990/1991, 2003/2004, 2004/2005, and 2021/2022 crop seasons.

In the 2021/2022 crop season, Brazil produced 125.5 million tons in 41.5 million ha. However, the yield loss in that year was 23.8%, equivalent to 30 million tons, or USD 16.3 billion. Higher yield losses were observed in MS (33.3%), PR (44.9%), SC (28.5%), and RS (58.3%). In this crop season, the yield loss observed in these four states was equivalent to 93.6% of the national yield loss, and the states in the South region alone represented 79% of the yield loss at the national level. The SPI corroborated this behavior when, from November 2021 to January 2022, a D4 (Exceptional Drought) event occurred, seriously compromising soybean production at regional and national levels.

It is noticeable, however, that drought occurrence in one specific important production region might undermine national production. As presented in Figure 6, the 2022/2023 crop season was a record-breaking soybean crop season with over 156 million tons produced in Brazil, and yield loss at the national level was 6.3% (9.4 million tons), mainly occurring in the RS State, representing 42.1% of state yield loss and 97% of national yield loss in that crop season. Hence, the SPI from September 2022 to February 2023 corroborated the yield losses related to drought observed in RS during the crop season (Figure 9).

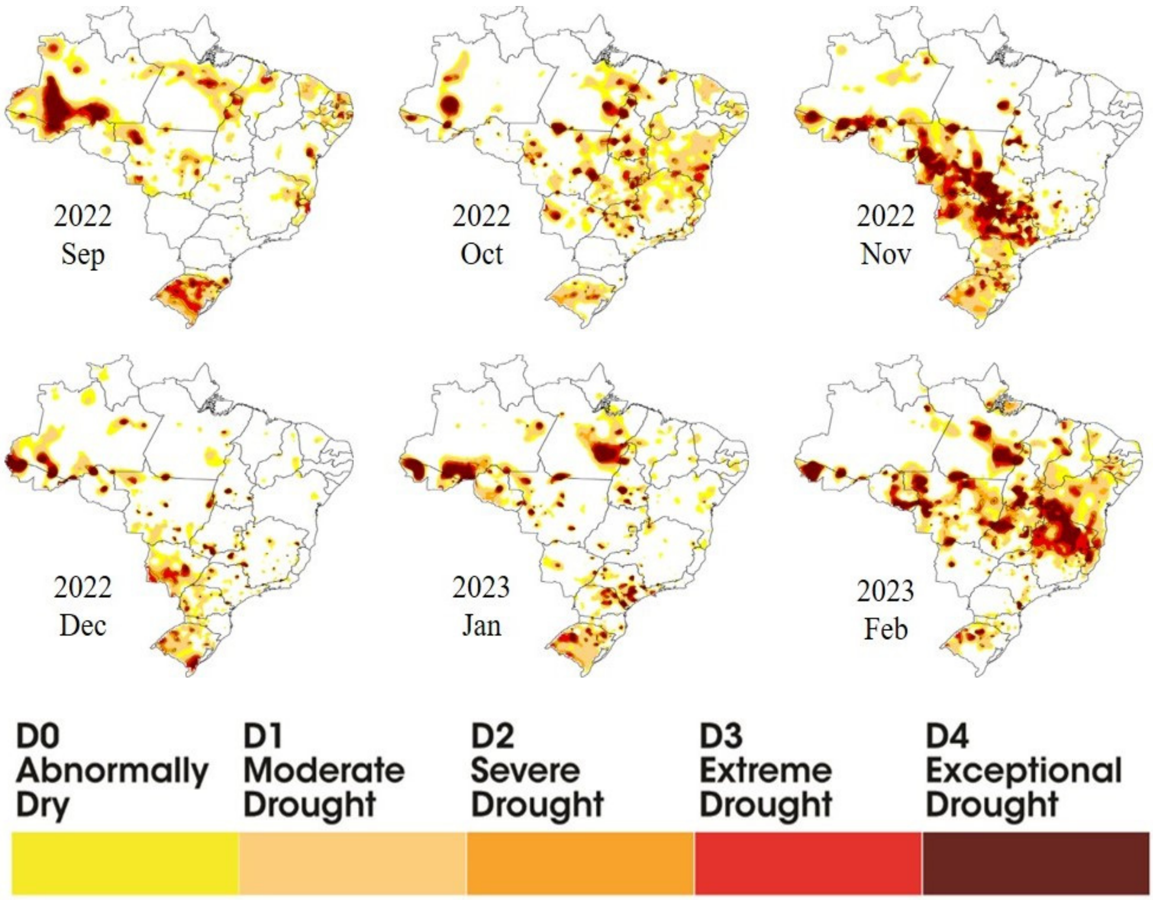

Figure 9. Monthly SPI maps from September, October, November, December, January, and February in the 2022/2023 crop season.

Figure 10 presents the occurrence of drought based on the SPI in the mesoregions of soybean production in Brazil in the last 20 crop seasons. Following the trend observed in yield loss at the state level (Figure 5a), it is possible to observe higher frequency in drought occurrence in the Central–West, Southeast, North, and Northeast regions compared to the South region. However, as previously discussed, although showing a lower drought frequency, the South region had more severe droughts, leading to higher yield losses than in preeminent soybean production regions in Brazil (Figure 5).

Splitting the 20-year SPI dataset into 10-year periods revealed alterations in the frequency of drought occurrence in the soybean production areas of Brazil. Specifically, increments in drought frequency were noticed in the Central–West, North, Northeast, and Southeast regions (and in the GO and MG States in the South region), and drought frequency demonstrated a reduction in the South region of Brazil. This change within two decades corroborates the trend of a lower frequency but higher severity of droughts and yield losses in the South region of Brazil compared to the Central–West region.

3.4. Spatial Distribution of Frequency and Severity of Soybean Yield Loss at the Municipal Level from 1974/1975 to 2021/2022 Crop Seasons

At the municipal level (Figure 11), the yield loss frequency is consonant with the trends observed at the state and national levels and with the SPI across the crop seasons.

Figure 11a evidences that yield loss occurrence at the municipal level is distributed evenly in Brazil. Among the 1998 municipalities analyzed (with at least 14 soybean crop seasons in the historical record), yield losses occurred in 68% (average) of the soybean crop seasons. In most municipalities, yield losses were observed in 50 to 80% of crop seasons. Figure 11b–d demonstrate that the yield loss at the municipal level follows the spread of soybean crops towards the Central–West region of Brazil and the MATOPIBA expansion area, proving that the frequency of yield from 1974/75 to 2003/2004 was similar to the entire historical record. This trend, with a high frequency of yield loss occurrence at the municipal level, remained equivalent from 2004/2005 to 2021/22 (Figure 11e,f), demonstrating that soybean yield loss in the majority of municipalities in Brazil is over 50%.

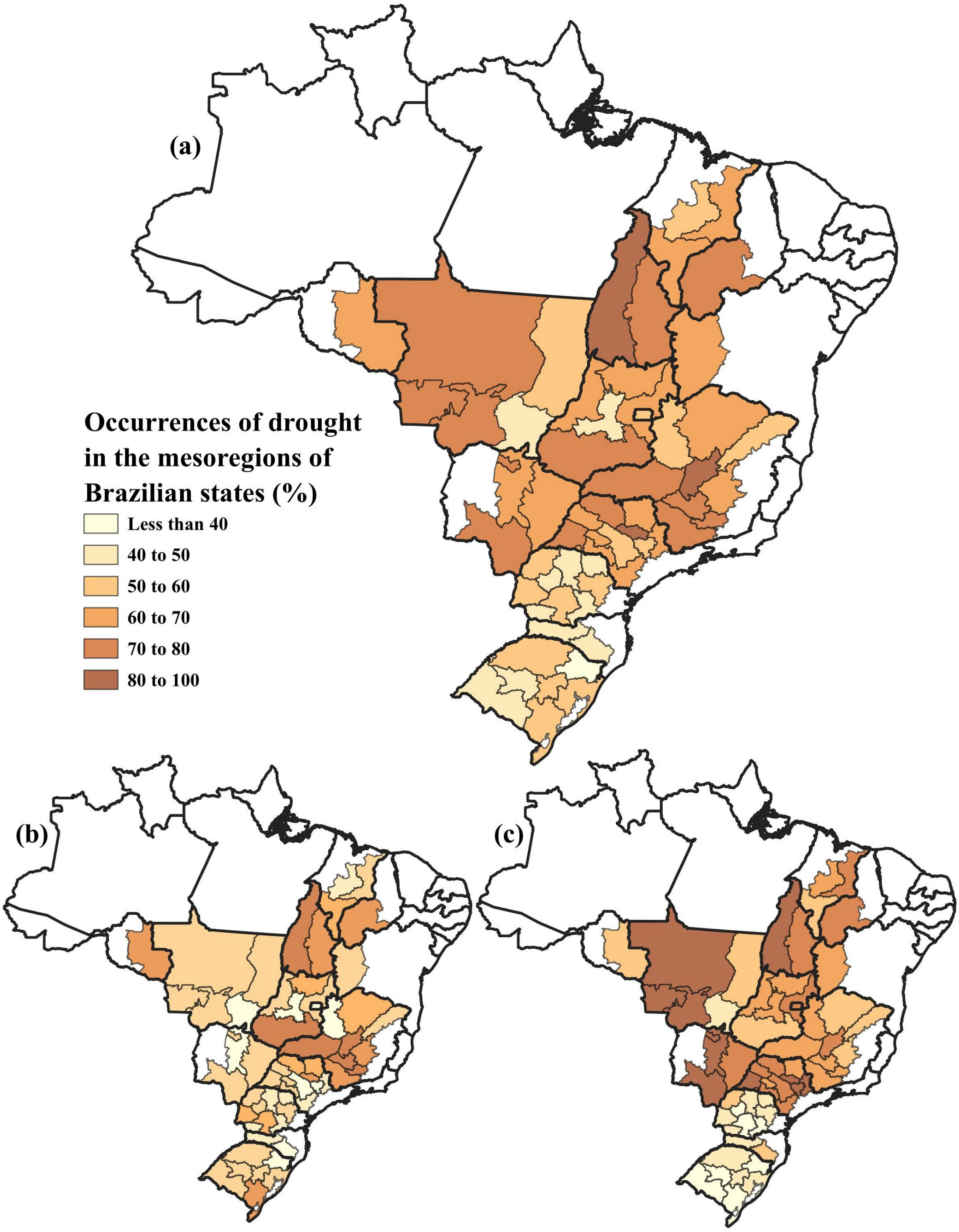

Figure 10. Drought occurrence based on SPI in the mesoregions of soybean production in Brazil from 2003/2004 to 2022/23 (**a**), from 2003/2004 to 2012/2013 (**b**), and from 2013/2014 to 2022/23 (**c**).

However, despite the even distribution of yield loss occurrence at the municipal level, with an average of 68% of crop seasons with soybean yield loss, the magnitude of loss is unevenly distributed. Figure 12 presents the frequency of soybean yield loss within classes of severity: from 0 to 10% (a), from 10 to 20% (b), from 20 to 30% (c), from 30 to 40% (d), and above 40% (e).

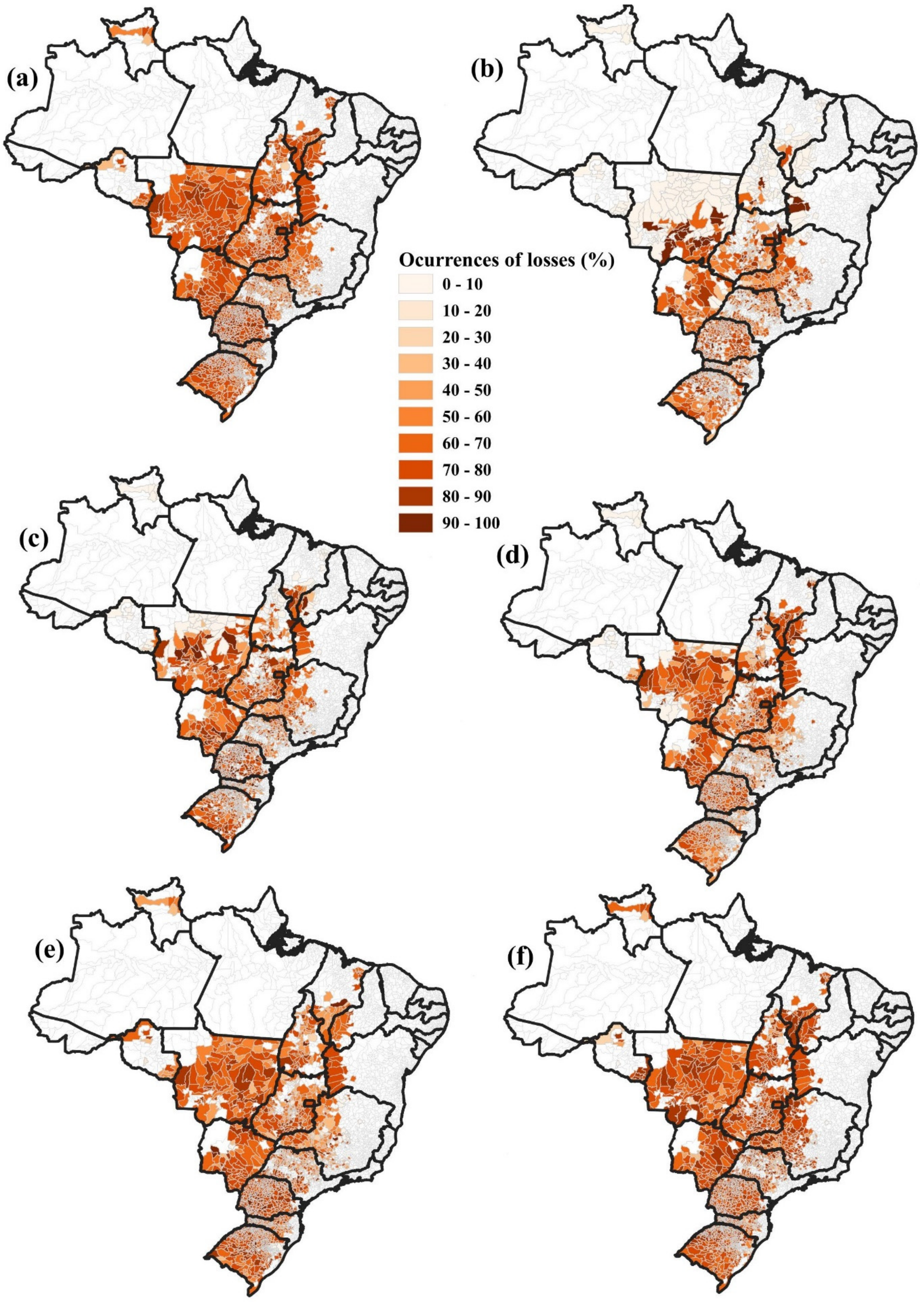

Figure 11. Occurrence of yield loss at the municipal level from 1974/1975 to 2020/2021 (**a**), from 1973/1974 to 1982/1983 (**b**), from 1983/1984 to 1992/1993 (**c**), from 1993/1994 to 2002/2003 (**d**), from 2003/2004 to 2013/2014 (**e**), and from 2014/2015 to 2020/2021 (**f**).

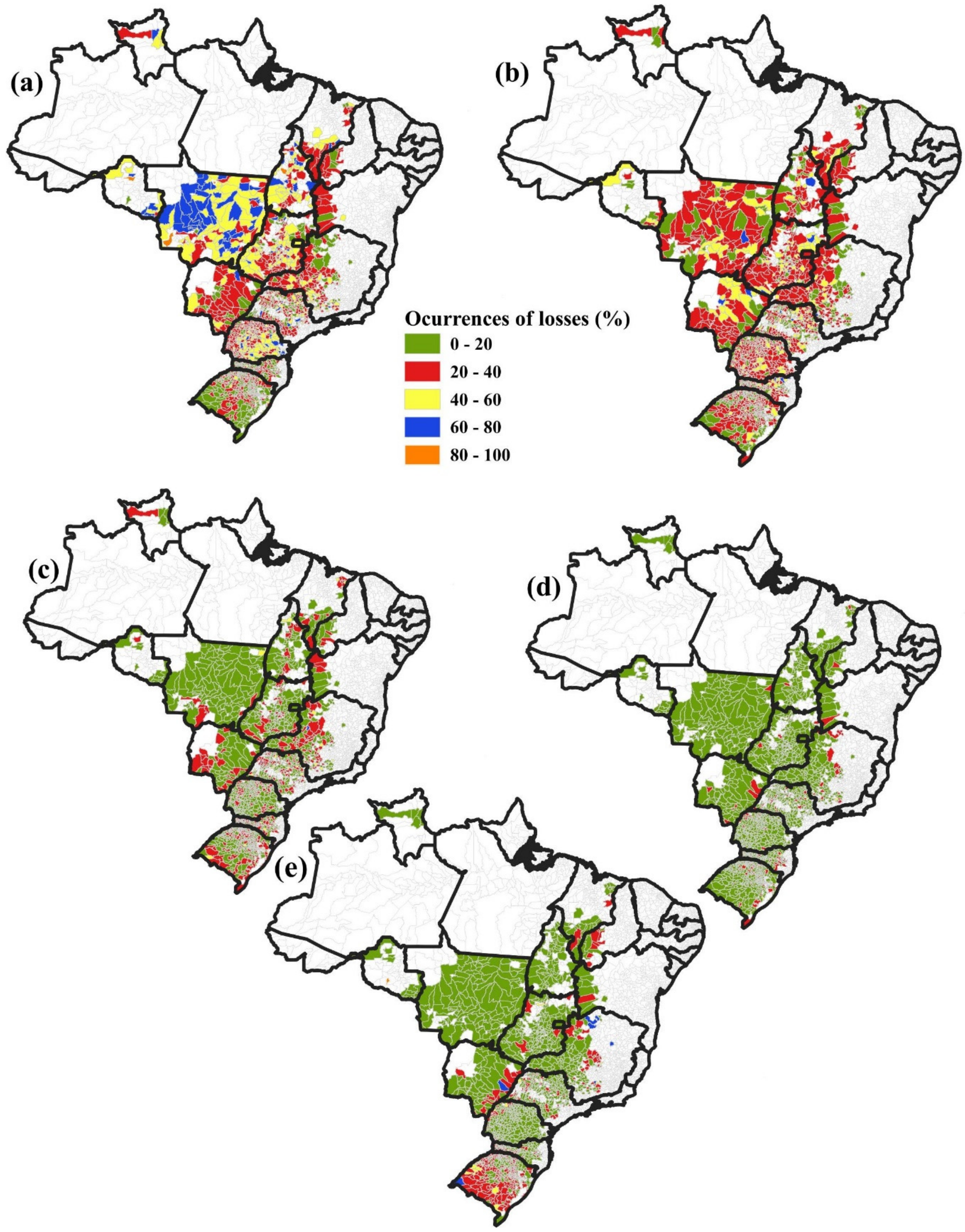

Figure 12. Severity of yield loss at the Municipal level expressed as a percentage of occurrence within severity levels: from 0 to 10% (**a**), from 10 to 20% (**b**), from 20 to 30% (**c**), from 30 to 40% (**d**), and above 40% (**e**).

From Figure 12a, it is possible to observe that yield loss occurrences from 0 to 10% at the municipal levels are concentrated in the central area of Brazil, specifically in municipalities located in MT State. An even yield loss distribution from 10 to 20% (Figure 11b) is evident in the 1998 municipalities with historical soybean production data in at least 14 crop seasons. As the severity of yield loss increases (Figure 12c,d), there is a trend of higher frequency of occurrence in municipalities located in GO, MS, MG, PR, SC, and RS States in addition to the MATOPIBA region.

Figure 12e highlights that municipalities with a high frequency of yield loss severity of over 40% are located most often in the RS State. Some specific municipalities located in the MS and MG States and the MATOPIBA expansion area presented a high frequency of yield loss above 40%. On average, about 25% of the crop seasons in municipalities in the RS State have a yield loss above 40%, against 14% of the municipalities outside of this state.

4. Discussion

4.1. Spatial–Temporal Trend in Soybean Yield Losses in Brazil

At the state and national levels, the temporal trend of soybean yield loss in Brazil demonstrated a similar frequency across the five decades, with losses in more than 50% of crop seasons, albeit with an observed decrease in severity, especially from 1990 onwards (Figures 5 and 6). Such reduction in severity at the national level might be related to improvements over the years in crop management practices. In turn, the spatial trend evidenced higher stability (lower losses) in the Central–West region, with the South region being more susceptible to yield losses, especially the RS State (Figures 5 and 7). Similar spatial–temporal trends at the municipal level (Figures 11 and 12) and yield losses could be verified using the SPI (Figures 8 and 9), evidencing drought occurrence in more than 40% of crop seasons and the tendency of increasing frequency in the last decade (Figure 10).

This spatial–temporal trend for soybean yield losses related to drought events led to an 11.65% impairment of Brazilian soybean production over five decades (Table 4). This cumulative amount represents 280 million tons, or USD 152 billion (Tables 2 and 3). Leng and Hall [27] created a global model of yield losses due to drought events on a long-term average record basis, establishing the probability of Brazilian and global losses. Their result was a loss of about 60% for both Brazilian and global production. However, our results reinforced that yield losses are distributed unevenly in Brazil and thus require local studies as well as global monitoring [21]; they also strengthen the contribution of a single climate index, such as the SPI, for spatial drought monitoring [28,29]).

4.2. Strategies to Mitigate the Effects of Drought on Soybean Yield

Based on the spatial and temporal trends in soybean yield losses related to drought events, it becomes crucial to determine how soybean production will develop in the upcoming years in Brazil. The future scenarios for climate variability reveal that it might be true that water availability will change across production areas. However, predicting the changes in precipitation indices is beyond the goals of the present research. Hence, considering that climate conditions are one of the few variables that cannot be fine-tuned for crop production, mitigation strategies will play an important role in the upcoming cropping season based on the spatial–temporal trends presented in Section 3.

Although irrigation contributes to increasing soybean yield stability in Brazil, providing adequate soil moisture levels during critical stages of crop development, only 800 thousand ha of Brazilian soybeans are under irrigation. This number represents 6% of all agricultural areas equipped with central-pivot irrigation and only 2% of all agricultural areas devoted to soybean production [5,42]. Even though irrigation might minimize yield losses under severe drought conditions [40], this is currently an unviable strategy for large-scale soybean production areas of Brazil in a short time interval.

Recent advances in biotechnology have shown high potential for developing soybean genotypes with higher tolerance to water-deficit periods, which would benefit soybean plants by helping equip them for longer periods with lower levels of water availability [16,43]. Thus, using more tolerant soybean cultivars might contribute to higher yield stability in the near future.

Meanwhile, considering the variability in time and space of drought events across soybean crop seasons, up-to-date information about crop development conditions would help the decision-makers in governmental and corporate institutions to cope with possible losses in soybean production. In this context, remote sensing technology has shown promise in the monitoring of water status in soybean crops and in forecasting yields [40]. These applications are potentially advantageous for adjustments in the trade market, financial projects, and food security policies.

To consolidate itself as the world's leading soybean producer, Brazil has made substantial investments in soybean production technologies, encompassing a variety of production systems in the country. The technologies used for soybean production in Brazil [7] and suggestions for crop management are summarized as follows: adequate plant population, observing seed quality; integrated disease and pest management; weed control; biological nitrogen fixation and soil fertility corrections; crop diversification in production systems; rational use of agrochemicals; and soil management practices to enlarge the soil water-holding capacity. These technologies are manageable and crucial to provide adequate conditions for crop development, thus minimizing the probability of physiological stress occurrence across crop seasons and promoting a higher capability of soybean plants to face water-deficit periods.

Based on the diversity of soybean production systems, Brazil classifies its production areas as edaphoclimatic regions with similar characteristics for the adaptability of soybean cultivars [39]. The parameters analyzed for grouping soybean production areas into five macroregions and twenty microregions are latitude (temperature and photoperiod), rainfall regime, altitude (temperature), and soil type. This public policy implemented by the Ministry of Agriculture, Livestock and Food Supply of Brazil (MAPA) in cooperation with the Brazilian Agriculture Research Corporation (Embrapa) annually indicates the best-adapted soybean cultivars according to their maturity groups for each production area in each region, contributing decisively to improving the stability of crop yields.

In addition, planting soybean cultivars with different maturity groups within the same production area (farm) and sowing the distinct sectors (plots within the farm) at different dates have the potential to avoid unfavorable climatic conditions in the entire production area (farm) simultaneously. This strategy provides different levels of susceptibility to drought occurrence and enlarges the possibility of overcoming negative impacts. At the regional level, adopting such a strategy might minimize losses in years of extreme drought conditions, which, in long-term projections, represents increments in crop yield. Additionally, considering the future climatic scenarios, Rio et al. [12] have proposed alternative sowing dates to mitigate the negative impacts on soybean yield due to drought occurrence.

4.3. Agricultural Zoning of Climatic Risk (ZARC) as Mandatory Public Policy for Agricultural Production, Farm Loan Programs, and Crop Insurance Contracts

The Agricultural Zoning of Climatic Risk (ZARC), established in 1996 by the Ministry of Agriculture, Livestock and Food Supply of Brazil (MAPA) in cooperation with the Brazilian Agriculture Research Corporation (Embrapa), has the ultimate goal of indicating what, where, and when to plant to minimize the risk of yield losses due to unfavorable climatic events. For soybean crops, ZARC classifies the risk of yield loss due to unfavorable climatic events, especially drought, in 10-day periods for all municipalities where MAPA indicates soybean cropping [44]].

ZARC is a well-established public policy in Brazil, which serves as a mandatory requirement for governmental and corporate policies related to agricultural production, farm loan programs, and crop insurance contracts. ZARC defines the risk derived from climatic variability based on the frequency of water-deficit occurrence during the most critical phases of soybean development (flowering and grain-filling stages) according to sowing windows, water availability in each region, crop water requirements on key phenological stages, soil water-holding capacity, and the maturity group of distinct cultivars [23]. As observed in Figure 6, the reduction in yield loss severity in the last 30 years (despite a similar frequency of yield loss) is deeply associated with ZARC, demonstrating its contribution to indicating more favorable sowing windows and minimizing soybean yield losses in Brazil.

Since 1996, ZARC for soybean crops has considered three soil types, mainly defined by their texture (clay content), to estimate the soil available water. From 2023, ZARC for soybean crops will adopt additional and more representative parameters to define six classes of soil available water considering clay, silt, and sand content based on pedotransfer functions representative of Brazilian soils [23].

In Brazil, the water source used to support crop water requirements and crop development is mostly precipitation (rainfall) and soil available water. Soil management practices might improve water storage, recharge, distribution, and deeper root systems, enlarging the available soil water. Such improvements might be crucial for providing higher soil moisture levels during water-deficit periods. Based on exposure, soil management practices play a role in improving hydric conditions, with direct influence to minimize yield losses due to drought conditions, the most limiting factor to yield stability in Brazil. Therefore, ZARC for soybean crops is currently developing an additional methodology to include the impact of different levels of soil management practices on soybean yield stability [45].

4.4. Current Limitations and Perspectives of the Adopted Methodology

Taking into account the tradeoff between spatial–temporal assessment of drought effects and the in situ crop conditions for yield loss monitoring, the limitations of the adopted methodology lie in the uncertainty of fully isolating the drought effect that impairs soybean yield. Thus, exploring the yield losses in the largest soybean producer in the world over five decades poses challenge in terms of acquiring in situ data on sowing date, used cultivar, soil and crop management practices, and meteorological conditions. It is particularly true that new diseases or insects may pose challenge to crop management. However, such variables occur in the totality of the area analyzed in the present research, diminishing their possible negative impact on the obtained results. Furthermore, other management practices are improved year by year, and there is no retrograded, leaving drought stress as the key factor responsible for year-to-year yield variability.

The limitations of the present investigation bring up the possibilities of developing a local-to-regional yield loss monitoring panel that gathers in situ information at the field scale. For that purpose, remote sensing technology might be used to map soybean fields and detect the sowing and harvesting date and also key phenological stages, in addition to contributing to the monitoring of crop rotation in the area. Agronomic information acquired from the farmers, like the cultivar planted and chemical defensives used, might provide insights into crop management. Soil analysis would also provide information on water-holding capacity and sustainable practices to enable greater stability of plants during water-deficit periods. Finally, climate information acquired both through satellite and weather stations (within a dense network) would provide climate information at the field level, enabling the estimation of yield losses based on the incorporation of the above mentioned variables.

5. Conclusions

The present research assessed the spatial–temporal trends in soybean yield losses related to drought events in Brazil over five decades at the national, state, and municipal levels and demonstrates yield losses in more than 50% of crop seasons at the national level, with similar frequency across the last five decades, albeit with observations of lower severities in the latest 30 years. Lower severities in the latter three decades represent a higher economic impact, given the increasing importance of soybean production to the national economy. Yield losses from 1976 to 1986 might be associated with the lack of specific management practices, especially in new agricultural frontiers. The Central–West region has proven more stable than the South region, where yield losses can frequently reach up to 74%. In five decades, yield losses related to drought events represent 11.65%, corresponding to 280 million tons, or USD 152 billion. The outcomes from the present research might subsidize public and corporate policies related to agricultural planning and zoning, farm loan programs, crop insurance contracts, and food security. The spatial–temporal assessment of yield loss will contribute to enlarging the sustainability of the soybean production chain, providing higher agricultural sustainability (expressed by lower losses and consequently higher yields), environmental sustainability (since higher yield stability prevents non-agricultural areas from being converted into soybean fields), economic sustainability (translated by greater profitability and public and corporate policies for the trade market, farm loan programs, and crop insurance contracts), and social sustainability (evidenced by food security in Brazil and globally).

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Data Availability Statement: The datasets used during the current study are available from the National Supply Company (CONAB), Brazilian Institute of Geography and Statistics (IBGE), and Brazilian Weather Forecast and Climate Studies Center from the National Institute of Space Research (CPTEC/INPE), as stated in the Section 2, and the dataset produced, processed, and analyzed in this paper is available from the corresponding author upon request.

Conflicts of Interest: Author Rodrigo Cornacini Ferreira was employed by the company Meta Agribusiness. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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